Large values and a zero-density estimate for the Riemann zeta function

A detailed study of a registered proof

An exposition of the development of Mark Lewko
with the analytic contributions credited in the original development

1 Introduction

The Riemann zeta function enters prime-number theory through its zeros. A zero-density theorem does not locate each zero individually. Instead, it bounds how many zeros can lie to the right of a given vertical line. This is one reason such theorems remain useful without the Riemann hypothesis: they permit a small exceptional set of zeros while still giving substantial information about the distribution of primes. Standard accounts of this connection and of zero-detecting Dirichlet polynomials are given by Montgomery [13], Ivić [8], and Iwaniec–Kowalski [9].

For $1/2 < \sigma < 1$ and $T \geq 2$, write

$$N(\sigma, T) = \sum_{\substack{\zeta(\rho)=0 \\ \operatorname{Re}\rho \geq \sigma,\ |\operatorname{Im}\rho| \leq T}} m(\rho),$$

where $m(\rho)$ is the order of vanishing of ζ at ρ. The finiteness of this sum and the conventions concerning its boundaries are established in Section 2. Throughout, zeros are counted with multiplicity and both signs of the ordinate are included. For $1/2 < \sigma < 1$, define the comparison exponent by

$$A(\sigma) = \inf\{a \geq 0 :\ N(\sigma, T) \ll_{\sigma,\varepsilon} T^{a(1-\sigma)+\varepsilon} \text{ for every } \varepsilon > 0\}.$$

We use it only to compare leading powers; our theorem retains explicit quantifiers and a positive power loss.

A central benchmark is the density hypothesis; see Montgomery [13], Titchmarsh [16], and the discussion in Tao–Trudgian–Yang [15]. We state it as background, not as a claim proved in this note.

Conjecture 1.1. *For every $1/2 < \sigma < 1$ and $\varepsilon > 0$,*

$$N(\sigma, T) \ll_{\sigma,\varepsilon} T^{2(1-\sigma)+\varepsilon} \qquad (T \geq 2).$$

The Riemann hypothesis would imply that the sum is empty for every $\sigma > 1/2$. Theorem 1.2 concerns only $\sigma > 3/4$ and does not establish either the Riemann hypothesis or the full Conjecture 1.1.

Historically, progress on zero density has been closely tied to progress on large values of Dirichlet polynomials. Ingham's estimate $A(\sigma) \leq 3/(2-\sigma)$ [7] is a classical landmark. The later theory combines mean-value inequalities, subdivision of the ordinate interval, estimates for correlations, and power-raising of Dirichlet polynomials; see [8, 9, 13]. Heath-Brown's twelfth-moment theorem [4] provides another important route: a zero detector that gives a

large value of a weighted critical-line mean can be controlled by a high moment. More recently, Guth–Maynard [3] obtained new large-values estimates near the three-quarter-power threshold, and Tao–Trudgian–Yang [15] organized many of the resulting exponent relations systematically.

The result studied here is the following theorem of the registered Lean development of Lewko and its credited contributors [11, 14]. It is stated first in the familiar mathematical form; Section 2 relates this statement to the exact zero sets and vanishing orders.

Theorem 1.2. *For every real $\sigma > 3/4$ and every $\varepsilon > 0$, there exists $C_{\sigma,\varepsilon} > 0$ such that*

$$N(\sigma, T) \le C_{\sigma,\varepsilon} T^{3(1-\sigma)/(2\sigma)+\varepsilon} \qquad (T \ge 2).$$

The case $\sigma \ge 1$ has zero count. The strict inequality $\sigma > 3/4$ is essential to this proof: the large-values argument needs a positive margin above the three-quarter-power threshold. No uniformity is claimed as $\sigma \downarrow 3/4$ or $\varepsilon \downarrow 0$.

It is important to distinguish this theorem's range from its exponent. The exponent $3/(2\sigma)$ is not new throughout the entire interval: Bourgain [2] obtained it for $\sigma \ge 3734/4694$, and Kerr [10, Theorem 6] extended that range to $\sigma \ge 23/29$. Relative to the bounds tabulated in Tao–Trudgian–Yang [15, Table 2], together with Kerr's preprint, the registered estimate is strictly smaller in leading exponent for $3/4 < \sigma < 23/29$, agrees with the best exponent located in those sources for $23/29 \le \sigma \le 7/8$, and is weaker than established estimates for $7/8 < \sigma < 1$. The last assertion already follows from the Heath-Brown bound

$$A(\sigma) \le \max\left\{\frac{3}{10\sigma - 7}, \frac{4}{4\sigma - 1}\right\},$$

recorded in [5, 15]. For $\sigma > 7/8$, both expressions are strictly smaller than $3/(2\sigma)$. These are comparisons with identified sources, not an exhaustive priority claim about all unpublished work. The present note makes no claim of a new result or of authorship of the underlying proof.

1.1 The two inputs and the proof strategy

The first input is the following large-values estimate, proved in Section 4. For a polynomial of length N, it bounds a one-separated set of ordinates at which the polynomial is large by

$$T^{\varepsilon}\left(N^2/V^2 + N^3 T^{1/2}/V^4\right), \qquad T^{1/3} \le N \le T, \quad V \ge N^{3/4+\kappa}.$$

The proof compares a complete pair sum with a smooth Poisson error. Poisson summation reflects each shell of ordinate differences to shorter frequency blocks. Mellin inversion separates the smooth weight from the oscillation without discarding any Mellin frequencies. A normalized supremum over all subsets and admissible scales then satisfies an inequality with a small coefficient times itself. This is the point at which $\kappa > 0$ is used; it makes that coefficient tend to zero uniformly.

The second input is the classical twelfth moment of Heath-Brown [4, Theorem 1]. The formulation needed here is

Theorem 1.3. *For every $\varepsilon > 0$, there are $K_\varepsilon > 0$ and T_0 such that*

$$\int_0^{3T} |\zeta(1/2 + \mathrm{i}t)|^{12}\,\mathrm{d}t \le K_\varepsilon T^{2+\varepsilon} \qquad (T \ge T_0).$$

Indeed, the classical estimate is $\int_0^U |\zeta(1/2+\mathrm{i}t)|^{12}\,\mathrm{d}t \ll U^2(\log U)^{17}$; put $U = 3T$ and absorb the logarithm into T^ε. For a first reading, this citation is a substantial and legitimate simplification. Appendix A and the subsequent appendices give the more detailed reconstruction used in the formal development: a fourth moment, pointwise control by a local mean, a signed stationary expansion, maximal-prefix estimates, and a high/low decomposition. That reconstruction is included for studying the methods, not to claim that the classical moment is an additional new theorem. Its relation to the classical work is discussed at the start of each appendix.

To connect these inputs to zeros, choose

$$a = \frac{1}{2\sigma}, \qquad c = \frac{3}{4\sigma}, \qquad p = \frac{3(1-\sigma)}{2\sigma}.$$

A short Möbius mollifier and an exponentially smoothed zero detector produce a large smooth dyadic block at every zero in $[T, 2T]$. If its length is at most $T^{c/2}$, expand the real-part-dependent weight in a binomial series. The coefficients of the resulting polynomial family are independent of the zero. Raising a selected member to a bounded integer power moves its length into $[T^a, 2^K T^c]$. The signs of the two length exponents in the large-values estimate then select opposite endpoints of this interval, and the identities

$$2(1-\sigma)c = \tfrac{1}{2} + (3-4\sigma)a = p$$

make both terms have the desired exponent.

The other zeros have a longer smooth block. Mellin inversion transfers it to a polynomially weighted critical-line mean. Weighted Hölder and a separated sampling estimate reduce its count to Theorem 1.3. The resulting exponent has a fixed saving because $c < 1$. Jensen's formula restores the multiplicities lost when choosing one-separated ordinates. Finally, dyadic summation and conjugation give Theorem 1.2, including the bounded-height part. Figure 1 displays these dependencies. Each arrow means that the estimate at its tail is used in the estimate at its head; it is not a graph of code imports.

1.2 Organization and conventions

Section 2 establishes the counting conventions. Section 3 recalls the analytic facts used in the main argument, and Section 4 proves the large-values estimate. Sections 5–7 construct the detector and prove the density theorem. The appendices provide a second, slower route through the moment argument and the analytic identities behind it.

All constants implicit in $\ll$ depend only on the fixed parameters explicitly indicated or fixed at the start of the relevant argument. An assertion holding for sufficiently large T, uniformly in a polynomial, a finite set or a zero, means that its threshold is chosen before those objects vary. Positive real powers are $x^u = \exp(u \log x)$; complex powers of positive real bases use the same real logarithm. Other branch-sensitive powers are principal powers. We use $s(t) = 1/2 + \mathrm{i}t$ and $Z(t) = |\zeta(s(t))|$; this Z is a nonnegative modulus, not Hardy's signed Z-function. The divisor function is $d(n) = \#\{r \in \mathbb{N} : r \geq 1,\ r \mid n\}$ for $n \geq 1$, with $d(0) = 0$ when a zero-indexed sum is used. We write $\gamma_E = \lim_{m\to\infty}(\sum_{j=1}^m j^{-1} - \log m)$ for Euler's constant, reserving γ for zero ordinates. All integrals over bounded intervals are Lebesgue integrals; oriented integrals retain their orientation. We use $\int_{(c)}$ for an upward vertical contour on $\operatorname{Re} s = c$.

This exposition is based on the immutable revision recorded in [11, 14], not on a moving default branch. The original mathematical and formal contributions, including the McColm and Grogan components credited upstream, remain attributed to their authors. This is an explanatory study, not a new formal certificate or an independently peer-reviewed proof.

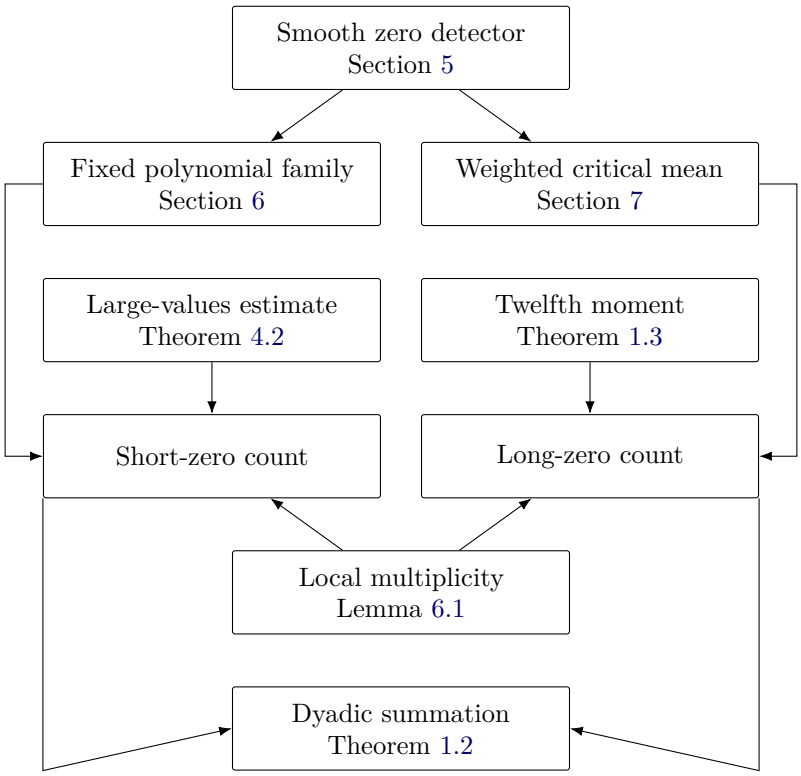


Figure 1: The two counting arguments and the common multiplicity step.

The original development uses mathlib [12]; the companion archive retains the argument pointers and licensing notices.

2 Actual zeros and their analytic multiplicities

Definition 2.1. A nontrivial zero is a complex number ρ satisfying

$$\zeta(\rho) = 0, \qquad 0 < \operatorname{Re}\rho < 1.$$

Define $m(\rho)$ by the local order of vanishing: it is k when $\zeta(z) = (z-\rho)^k g(z)$ near ρ with $g(\rho) \neq 0$; it is zero at a nonzero analytic value. An identically zero germ has infinite extended order. The next lemma excludes that case for $\rho \neq 1$, so $m(\rho)$ is a natural number at every zero used here. Write

$$\begin{aligned} \mathcal{Z}(\sigma; a, b) &= \{\rho : \rho \text{ is a nontrivial zero},\ \sigma \leq \operatorname{Re}\rho,\ a \leq \operatorname{Im}\rho \leq b\}, \\ N_{\mathrm{slab}}(\sigma; a, b) &= \sum_{\rho \in \mathcal{Z}(\sigma;a,b)} m(\rho), \\ N(\sigma, T) &= N_{\mathrm{slab}}(\sigma; -T, T). \end{aligned}$$

Finiteness and local factorization.

Lemma 2.2. *The set $\mathcal{Z}(\sigma; a, b)$ is finite for arbitrary real σ, a, b. At every nontrivial zero, $0 < m(\rho) < \infty$, and there is a function g analytic at ρ, with $g(\rho) \neq 0$, for which*

$$\zeta(z) = (z-\rho)^{m(\rho)} g(z)$$

in a neighborhood of ρ.

Proof. The zeta function is meromorphic with its only pole at 1, and $F(z) = (z-1)\zeta(z)$ extends holomorphically across 1, with $F(1) = 1$. Thus F is entire and not identically zero. Its zeros in any compact set are finite: otherwise a convergent subsequence would have an accumulation point, contradicting the identity theorem. Every zeta zero in the region below is also a zero of F. This justifies using a closed ball even if it contains the pole of ζ. Every zero in the displayed region has

$$|\rho| \le |\operatorname{Re}\rho| + |\operatorname{Im}\rho| \le 1 + |a| + |b|.$$

Thus the region is a subset of the zeros in a compact closed ball, and is finite.

If the analytic order at some $\rho \neq 1$ were infinite, the analytic germ would vanish identically near ρ. The identity principle on the connected punctured plane would then make ζ identically zero on $\mathbb{C} \setminus \{1\}$, contradicting $\zeta(2) \neq 0$. Hence its extended order is finite. Analyticity and $\zeta(\rho) = 0$ give a nonzero order; the ordinary order of vanishing is therefore a positive integer. The local factorization theorem then gives g and the displayed local identity. A nontrivial zero is not 1, since its real part is strictly less than 1, so all these facts apply. □

Exact membership at the endpoint.

Lemma 2.3. *If $\sigma > 3/4$, then*

$$\rho \in \mathcal{Z}(\sigma; -T, T) \iff \zeta(\rho) = 0 \text{ and } \sigma \le \operatorname{Re}\rho \text{ and } |\operatorname{Im}\rho| \le T.$$

If $\sigma \ge 1$, the set is empty for every T.

Proof. One implication simply forgets the strip conditions. Conversely, a zero with $\operatorname{Re}\rho \ge \sigma > 3/4$ has positive real part. Mathlib's nonvanishing theorem for $\operatorname{Re} s \ge 1$ forces $\operatorname{Re}\rho < 1$. Thus it is a nontrivial zero, and $-T \le \operatorname{Im}\rho \le T$ is precisely $|\operatorname{Im}\rho| \le T$. If $\sigma \ge 1$, membership would give both $\sigma \le \operatorname{Re}\rho$ and $\operatorname{Re}\rho < 1$, an impossibility. □

Monotonicity and conjugation. The slab count increases when either the region is enlarged by lowering the real-part cutoff, or its lower height is lowered, or its upper height is raised. This follows by inclusion of the finite zero sets and nonnegativity of multiplicities. Complex conjugation preserves zeros, real part and analytic order, and reverses ordinate. Consequently

$$N(\sigma, T) \le 2N_{\text{slab}}(\sigma; 0, T).$$

The inequality, rather than an equality, permits zeros on the real axis and closed shared boundaries. Similarly

$$N_{\text{slab}}(\sigma; a, c) \le N_{\text{slab}}(\sigma; a, b) + N_{\text{slab}}(\sigma; b, c) \qquad (a \le b \le c).$$

These finite-set statements are what the final dyadic argument uses.

3 Analytic preliminaries

We use the functional equation and conjugation of the Riemann zeta function, its nonvanishing on $\operatorname{Re} s \ge 1$, Mellin inversion of the exponential, Poisson summation for smooth compactly supported functions, Jensen's formula, and the elementary estimate $d(n) \ll_\theta n^\theta$ for every $\theta > 0$. These facts are used in their usual domains of convergence; infinite interchanges below are justified by absolute summability or an integrable majorant. For reference, see [8, 9, 16]. The Fourier and Mellin conventions are

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x\xi}\, \mathrm{d}x, \qquad \mathcal{M}f(s) = \int_0^\infty f(x) x^{s-1}\, \mathrm{d}x.$$

With $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$, the completed zeta is $\Lambda(s) = \Gamma_{\mathbb{R}}(s)\zeta(s)$ away from its poles and satisfies $\Lambda(s) = \Lambda(1-s)$. Since $s(t)$ never meets the pole, Z is continuous and every natural power Z^n is integrable on each bounded interval. No global moment estimate follows merely from this local integrability.

The pointwise estimate needed by the zero detector is the classical Weyl bound, in the following power-loss formulation [8, 16].

Proposition 3.1. *For each $\theta > 0$, there is $C_\theta > 0$ such that*

$$Z(t) \le C_\theta(1 + |t|)^{1/6+\theta} \qquad (t \in \mathbb{R}).$$

The finite-sum proof and the sharp approximate functional equation used in the registered development are reconstructed in Appendix F. Applying the bound with loss $\theta/2$ and absorbing a fixed factor of $2^{1/6+\theta/2}$ gives, uniformly on $[H, 2H]$, $Z(t) < H^{1/6+\theta}$ for sufficiently large H. We also use the coarser bound $Z(t) \le 4(1 + |t|)$ obtained by Abel continuation in the development. Polynomial weights with sufficiently large decay and exponential weights consequently give genuine convergent integrals. In particular $Z(t)^{12} \le H^3$ on $[H, 2H]$ for sufficiently large H; this is only an upper truncation, not a moment theorem.

4 A large-values estimate for Dirichlet polynomials

Large-value data.

Definition 4.1. For positive integer N, coefficients $a_n \in \mathbb{C}$, and real t, put

$$D_{N,a}(t) = \sum_{N<n\le 2N} a_n e^{it\log(n/N)}.$$

A finite real set W is one-separated if $|t-u| \ge 1$ whenever $t, u \in W$ and $t \ne u$. For fixed $\kappa > 0$, admissible data are T, V, N, a, W, t_0 with

$$T \ge 2, \quad N \ge 1, \quad V > 0, \quad T^{1/3} \le N \le T, \quad N^{3/4+\kappa} \le V,$$
$$|a_n| \le 1 \ (N < n \le 2N), \quad W \subset [t_0, t_0 + T], \quad W \text{ one-separated}, \quad |D_{N,a}(t)| \ge V \ (t \in W).$$

For a finite nonempty U set $d_U = 1 + \max U - \min U$, and put $d_\varnothing = 1$. Finally define

$$\Delta(T) = \max_{1\le m\le \lceil 32(T+1)\rceil} d(m), \qquad b = \frac{N\Delta(T)}{V^2}.$$

A large-values estimate.

Theorem 4.2. *For every $\kappa, \varepsilon > 0$ there is $C > 0$, chosen before every admissible data set, such that*

$$|W| \le CT^\varepsilon\left(\frac{N^2}{V^2} + \frac{N^3\sqrt{T}}{V^4}\right).$$

The starting point t_0 is arbitrary.

4.1 Complete pair sums and a smooth comparison

Choose once and for all a nonnegative C^∞ cutoff w equal to 1 on $[1,4]$ and supported in $[1/2,5]$. We construct it as a smooth bump centered at $5/2$, inner radius $3/2$, outer radius 2; its actual outer support $[1/2,9/2]$ is contained in the stated larger interval. Smoothness and support are used for reflection; nonnegativity and the plateau are used for comparison. Define

$$\begin{aligned} I_w(v) &= \int_0^\infty w(x)x^{\mathrm{i}v}\,\mathrm{d}x, \\ h_w(L,v) &= \sum_{n\ge 0} w(n/L)(n/L)^{\mathrm{i}v} - LI_w(v), \\ P_M(v) &= \sum_{M\le q<2M} (q/M)^{\mathrm{i}v}, \\ E_M(U) &= \sum_{t,u\in U} |P_M(t-u)|^2, \\ Q_L(U) &= \sum_{t,u\in U} |h_w(L,t-u)|^2. \end{aligned}$$

The apparently infinite discrete sum defining h_w has finite support, and its $n=0$ term is zero. Notice the distinct half-open conventions: the auxiliary polynomial uses $[M,2M)$; the detecting polynomial uses $(N,2N]$.

Comparison with the kernel energy.

Lemma 4.3. *There is $C_w \ge 0$ such that for all admissible data, every one-separated subset $U \subset W$, and every positive integer M with $NM \le 8(T+1)$,*

$$E_M(U) \le C_w bNM^2|U| + b^2 Q_{NM}(U).$$

Also $E_M(U) \ge M|U|^2$.

Proof. Expanding the complete pair energy in the q,r variables gives its Gram form

$$E_M(U) = \sum_{M\le q,r<2M} \left|\sum_{t\in U} (q/r)^{\mathrm{i}t}\right|^2.$$

The diagonal $q=r$ contains exactly M copies of $|U|^2$, proving the lower bound. For the upper bound, insert $D_{N,a}(t)/D_{N,a}(t)$ inside each t sum. These denominators are nonzero by $|D_{N,a}(t)| \ge V > 0$. Finite Cauchy–Schwarz and $|a_n| \le 1$ give the factor N and sums over products $m = nq$. A fixed product has at most $d(m)$ representations, bounded by $\Delta(T)$ because the scale restriction places it within the defining maximum. The cutoff equals one on the original normalized product support, and is nonnegative, so extending to all cutoff-supported integers does not decrease the sum.

The resulting extension, when its squares are expanded in t,u, contains

$$NMI_w(t-u) + h_w(NM, t-u)$$

multiplied by $\overline{P_M(t-u)}$ and divided by $D_{N,a}(t)\overline{D_{N,a}(u)}$. More explicitly, with $D(t) = D_{N,a}(t)$, the product grouping gives

$$E_M(U) \le N\Delta(T) \sum_{M\le r<2M} \sum_{m\ge 0} w(m/(NM)) \left|\sum_{t\in U} \frac{(m/(NM))^{\mathrm{i}t}(r/M)^{-\mathrm{i}t}}{D(t)}\right|^2.$$

The zero-indexed term is zero because the cutoff vanishes near zero. On the original product range $m = nq$, $w(m/(NM)) = 1$. After expansion of the square and summation over r, the preceding inequality contains exactly the continuous and error terms displayed above. Taking their absolute values and using $|D(t)D(u)| \geq V^2$ gives the scalar $b = N\Delta(T)/V^2$. The continuous piece has $|I_w(v)| \leq C(1+|v|)^{-2}$: set $x = e^{-y}$ and apply Fourier decay to the compact smooth logarithmic lift. One-separatedness gives a uniform row sum

$$\sum_{u \in U} (1 + |t - u|)^{-2} \leq S$$

for a fixed absolute constant S. One may obtain such a constant by dividing each side of t into half-open unit intervals. Each contains at most one point, and the resulting row sum is bounded by twice a convergent sum of $(1+j)^{-2}$. Thus the continuous piece is at most a fixed multiple of $bNM^2|U|$. For the remaining piece use, pair by pair,

$$b|h_w(NM, t-u)|\,|P_M(t-u)| \leq \tfrac{1}{2}b^2|h_w(NM, t-u)|^2 + \tfrac{1}{2}|P_M(t-u)|^2.$$

After summing, the last term is $E_M(U)/2$ and is moved to the left. This produces the displayed comparison, with its constant chosen from w alone. □

4.2 Poisson summation and Mellin separation

For $H, L > 0$ the retained integer band is exactly

$$\mathcal{B}(L, H) = \left\{ m \in \mathbb{Z} : m > 0, \quad \frac{H}{20\pi L} \leq m \leq \frac{4H}{\pi L} \right\}.$$

The mode integral is

$$J_w(v, k) = \int_0^\infty w(x) e^{\mathrm{i}(v \log x - 2\pi k x)}\, \mathrm{d}x.$$

Poisson summation gives the absolutely convergent identity

$$h_w(L, v) = L \sum_{m \neq 0} J_w(v, mL).$$

For $H \leq v \leq 2H$, the off-band sum has norm at most $C_A L^{1-A}$, for every integer $A > 1$. The same estimate holds for $h_w(L, 0)$. To prove it, integrate repeatedly by parts with quotient $x/(v - 2\pi k x)$; off-band the denominator has the explicit lower bound $(H + |k|)/300$ on $[1/3, 6]$. Rescaling puts parameters in a compact set, so all iterated amplitudes have uniform bounds. Summing $|m|^{-A}$ justifies the discarded-mode estimate. Endpoint amplitudes vanish because the support lies strictly within this interval.

All-frequency oscillatory estimate.

Lemma 4.4. *For every real τ and $0 < a \leq b$,*

$$\left| \int_a^b z^{-1} e^{\mathrm{i}(\tau \log z - 2\pi z)}\, \mathrm{d}z \right| \leq \frac{6}{\sqrt{2\pi a}}.$$

Proof. Put $s = \sqrt{2\pi a}$ and $L_0 = (\tau - s)/(2\pi)$, $R_0 = (\tau + s)/(2\pi)$. Away from the stationary point, write the integrand as

$$\frac{1}{\mathrm{i}(\tau - 2\pi z)} \frac{\mathrm{d}}{\mathrm{d}z} e^{\mathrm{i}(\tau \log z - 2\pi z)}.$$

Integration by parts bounds an interval to the left of L_0 or to the right of R_0 by $2/s$, since the reciprocal denominator is monotone there and has modulus at most $1/s$ at the closest endpoint. If the original interval lies wholly on one side, this already suffices. Otherwise split it at $u = \max(a, L_0)$ and $v = \min(b, R_0)$. Each outside part is bounded by $2/s$, with zero-length parts contributing zero. On $[u, v]$, absolute integration gives

$$\frac{v-u}{a} \le \frac{R_0 - L_0}{a} = \frac{2}{s},$$

using $s^2 = 2\pi a$. The three estimates sum to $6/s$. This argument makes no assumption on the sign of τ. □

For $m \in \mathcal{B}(L, H)$, change variables $z = mLx$ in $J_w(v, mL)$ and apply Mellin inversion on the line $\operatorname{Re} s = 1$. All retained modes use the same interval $[H/(40\pi), 20H/\pi]$. Thus

$$J_w(v, mL) = \frac{1}{2\pi} \int_{\mathbb{R}} \mathcal{M}w(1 + \mathrm{i}r)(mL)^{\mathrm{i}(r-v)} K(H, v, r)\, \mathrm{d}r,$$

where

$$K(H, v, r) = \int_{H/(40\pi)}^{20H/\pi} z^{-1} e^{\mathrm{i}((v-r)\log z - 2\pi z)}\, \mathrm{d}z.$$

Lemma 4.4 gives $|K(H, v, r)|^2 \le 2000/H$ for *every* r. Weighted Cauchy–Schwarz, with weight $|\mathcal{M}w(1+\mathrm{i}r)|$, then bounds the retained band by a squared short phase sum. Its Mellin mass is finite by the compact-smooth Fourier decay. There is no discarded large-r region and no hidden restriction $|r| \ll H$.

For a shell $H \le |t-u| < 2H$, both signs are handled by conjugation of the real cutoff. The retained integer band is partitioned into at most ten dyadic blocks. The phase coefficients on each block have modulus one, and the complete-pair Gram expansion dominates coefficients of modulus at most one by the unweighted complete-pair energy. Localize U to

$$U_{H,j} = U \cap [jH, (j+3)H), \qquad j \in \mathbb{Z}.$$

Each point belongs to exactly three such sets; their diameters are at most $3H$, hence their local diameters at most $4H$ when $H \ge 1$. Write $\mathscr{R}(L, H; U)$ for the exact finite energy

$$\mathscr{R}(L, H; U) = \sum_{M \in \mathcal{B}_{\mathrm{dyad}}(L,H)} \sum_{j \in \mathcal{J}(U,H)} E_M(U_{H,j}),$$

where $\mathcal{B}_{\mathrm{dyad}}(L, H) = \{2^{\lfloor \log_2 m \rfloor} : m \in \mathcal{B}(L, H)\}$ and $\mathcal{J}(U, H) = \bigcup_{t \in U} \{\lfloor t/H \rfloor - 2, \dots, \lfloor t/H \rfloor\}$. The latter is exactly the finite set of nonempty local-window indices. The band block lengths range from $H/(40\pi L)$ to $4H/(\pi L)$; the ratio is at most 160, which places all their exponents in an interval of at most ten integers. This proves the stated finite block bound. The exact reflection estimate is

$$\sum_{\substack{t,u \in U \\ H \le |t-u| < 2H}} |h_w(L, t-u)|^2 \le \frac{20000}{\pi^2} \left(\int_{\mathbb{R}} |\mathcal{M}w(1+\mathrm{i}r)|\, \mathrm{d}r \right)^2 \frac{L^2}{H} \mathscr{R}(L, H; U) + 4C_A^2 |U|^2 L^{2-2A}. \tag{4.1}$$

The diagonal has the separate bound $C_A^2 |U| L^{2-2A}$.

4.3 A normalized supremum and uniform absorption

For fixed nonempty admissible data define the finite-valued supremum

$$B = \sup_{\substack{\varnothing \ne U \subset W \\ N \le L \le 8(T+1)}} \frac{Q_L(U)}{b|U|(L^2 + Nd_U)}.$$

This is not an assumed kernel bound. The family is nonempty. Since $V \le N$ for nonempty W, $\Delta(T) \ge 1$ and $L \ge N$, the denominator is at least 1. A crude finite-support bound on $h_w(L, v)$, uniform for $0 < L \le 8(T+1)$, bounds each numerator by a fixed multiple of $|W|^2$. Therefore the supremum is finite and nonnegative, and every descendant kernel is bounded by its defining normalization times B.

Bootstrap inequality.

Lemma 4.5. *There are constants $C_0, C_1 \ge 0$ depending only on w such that*

$$B \le C_0 + C_1 N b^2 \log(2T) B.$$

Proof. Apply the comparison lemma to every reflected dyadic block and every local set in (4.1). For a reflected positive integer dyadic length M', the band gives $M' \le 4H/(\pi L)$. Since $L \ge N$ and $H \le T$, we have $NM' \le 4NH/(\pi L) \le 4T/\pi \le 8(T+1)$, while $NM' \ge N$ because $M' \ge 1$. Thus the comparison kernels remain within the supremum's scale range. For a reflected dyadic length M, the band geometry gives $(L^2/H)M^2 \le 16H$. The total local-cardinality sum is $3|U|$ and there are at most ten reflected blocks. Consequently each shell is bounded by

$$CbNH|U| + CBb^3|U|(N^2H + NL^2) + ET^{-100}.$$

We obtain the error by taking a sufficiently high integration-by-parts order and using $L \ge N \ge T^{1/3}$, $|U| \le T+1$. Explicitly the off-band recursion uses $R_0 = H + |k|$ and normalized $p = (v/R_0, 2\pi k/R_0)$ in the compact set

$$|p_1| \le 2, \quad |p_2| \le 2\pi, \quad |p_1 - p_2 x| \ge 1/300 \quad (1/3 \le x \le 6).$$

Define $a_0 = w$ and $a_{j+1}(x) = \mathrm{d}[a_j(x)x/(p_1 - p_2 x)]/\mathrm{d}x$. All supports remain in $[1/2, 5]$. Joint smoothness on this compact parameter set bounds a_j uniformly. If I_j is its integral against $e^{\mathrm{i}R_0(p_1 \log x - p_2 x)}$, integration by parts gives $R_0 \mathrm{i} I_j = -I_{j+1}$, with zero boundary terms. Thus $|I_0| \le 6C_j/R_0^j$. Summing its integer-mode majorants and choosing j large enough yields the stated shell error before any supremum absorption. The one-separated pairs have no nonzero gap below 1 and no gap exceeding $\operatorname{diam} U$. Summing dyadic gap shells gives $\sum H \le 2d_U$ and at most $\log(2T)/\log 2$ shells. With the diagonal added,

$$Q_L(U) \le 2CbNd_U|U| + CBb^3|U|\left(2N^2 d_U + NL^2 \frac{\log(2T)}{\log 2}\right) + E'T^{-90}.$$

Divide by $b|U|(L^2 + Nd_U)$, whose lower bound also absorbs the last error, and take the supremum. The first normalized term is bounded by a constant; the recursive term is at most a constant times $Nb^2 \log(2T)B$. This proves the first assertion without any circular assumption.

To extract its uniform consequence, use the height condition:

$$Nb^2 = \frac{N^3 \Delta(T)^2}{V^4} \le \Delta(T)^2 N^{-4\kappa}.$$

Use $\Delta(T) \le D_\kappa T^{\kappa/6}$ and $N \ge T^{1/3}$ to obtain $Nb^2 \le D_\kappa^2 T^{-\kappa}$. Both this expression and its product with $\log(2T)$ tend to zero. A single threshold depending on κ, w therefore makes $Nb^2 \le 1$ and $C_1 Nb^2 \log(2T) \le 1/2$ for every admissible configuration. Since $B \ge 0$, the inequality implies $B \le C_0 + B/2$, hence $B \le 2C_0$. □

We now prove Theorem 4.2.

Proof. The absorbed kernel satisfies $Q_L(W) \le Kb|W|(L^2 + Nd_W)$ uniformly for $N \le L \le 8(T+1)$. Combine this at $L = NM$ with Lemma 4.3 and $E_M(W) \ge M|W|^2$. Divide by $M|W|$ for nonempty W and use $Nb^2 \le 1$:

$$|W| \le (C_w + K)bNM + Kb^3\frac{Nd_W}{M}.$$

Choose the positive integer $M = \lceil b\sqrt{T}\rceil$. Its rounding bounds $b\sqrt{T} \le M \le b\sqrt{T}+1$ and $N \le T$ show that $N \le NM \le 8(T+1)$. The argument explicitly derives $Nb\sqrt{T} \le T$ from $Nb^2 \le 1$ and $N \le T$. Furthermore,

$$bNM \le Nb + Nb^2\sqrt{T}, \qquad b^3\frac{Nd_W}{M} \le 2Nb^2\sqrt{T},$$

since $d_W \le T+1 \le 2T$. This gives, above the uniform threshold,

$$|W| \le A\left(\Delta(T)\frac{N^2}{V^2} + \Delta(T)^2\frac{N^3\sqrt{T}}{V^4}\right).$$

The divisor maximum estimate with loss $\varepsilon/2$ bounds both $\Delta(T)$ and $\Delta(T)^2$ by D^2T^{ε}, with $D \ge 1$. For heights below the threshold, one-separatedness gives $|W| \le T+1$. If $W \ne \varnothing$, $V \le N$, so the bracket $N^2/V^2 + N^3\sqrt{T}/V^4$ is at least one, and $T^{\varepsilon} \ge 1$. Adding the bounded-height threshold and 1 to the large-height constant therefore handles every $T \ge 2$. The empty set has zero cardinality and satisfies the same inequality immediately. □

5 A smooth zero detector

We now reconstruct the zero-density application, using the analytic estimates already proved. Fix $3/4 < \sigma < 1$, and write

$$a = \frac{1}{2\sigma}, \qquad c = \frac{3}{4\sigma} = \frac{3}{2}a, \qquad p = \frac{3(1-\sigma)}{2\sigma}, \qquad X = T^{\delta}, \quad Y = T^{c}. \tag{5.1}$$

Then $1/2 < a < 2/3$, $0 < c < 1$, $p > 0$. Choose $0 < \delta \le 1/8$ with $\delta < (1-c)/12$ before T and any zero. One may take, for example, $\delta = \min(1/16, (1-c)/24)$. The exact identities used later are

$$2(1-\sigma)c = p, \qquad \tfrac{1}{2} + (3-4\sigma)a = p, \qquad 2 + (3-6\sigma)c = p - (1-c). \tag{5.2}$$

They follow by substituting (5.1) and clearing the positive σ denominator. They are balances of the two short terms and the long saving, not asymptotic approximations.

Möbius mollifier and detector coefficients.

Definition 5.1. Let μ denote the Möbius function: $\mu(1) = 1$, $\mu(n) = 0$ if n is divisible by a prime square, and $\mu(n) = (-1)^r$ if n is a product of r distinct primes. For a real X, define

$$M_X(s) = \sum_{1 \le d \le \lfloor X\rfloor_+} \mu(d)d^{-s}, \qquad b_X(n) = \sum_{\substack{d \mid n \\ d \le \lfloor X\rfloor_+}} \mu(d), \qquad b_X(0) = 0.$$

Here $\lfloor X\rfloor_+$ is the natural floor used by Lean; for $X \ge 0$ it is the ordinary natural floor. Put

$$\mathcal{D}(\rho; X, Y) = \sum_{n \ge 1} b_X(n)n^{-\rho}e^{-n/Y}, \qquad \mathcal{T}(\rho; X, Y) = \sum_{n \ge 2} b_X(n)n^{-\rho}e^{-n/Y}.$$

If $X \geq 1$, then $b_X(1) = 1$, $b_X(n) = 0$ for $2 \leq n \leq X$, and $|b_X(n)| \leq d(n)$. These follow from the exact Möbius divisor cancellation and $|\mu(d)| \leq 1$. Also $|M_X(1/2 + \mathrm{i}t)| \leq 2\sqrt{X}$ for $X \geq 0$, by the finite $n^{-1/2}$ sum comparison.

Exact exponentially smoothed contour identity.

Lemma 5.2. *Let $\rho = \beta + \mathrm{i}\gamma$ be a zero with $3/4 \leq \beta < 1$, and $Y \geq 1$. The detector series is absolutely convergent and*

$$\mathcal{D}(\rho; X, Y) = M_X(1)Y^{1-\rho}\Gamma(1-\rho) + \frac{1}{2\pi}\int_{\mathbb{R}} \Gamma(1/2 - \beta + \mathrm{i}u)\zeta(1/2 + \mathrm{i}(\gamma+u))M_X(1/2 + \mathrm{i}(\gamma+u))Y^{1/2-\beta+\mathrm{i}u}\,\mathrm{d}u. \tag{5.3}$$

Proof. Mellin inversion of $e^{-n/Y}$ on the initial right line $\operatorname{Re} w = 1/2$ gives the integral of $\Gamma(w)\zeta(\rho + w)M_X(\rho + w)Y^w$. This line has $\operatorname{Re}(\rho + w) > 1$, where the Dirichlet product equals the series with coefficients $b_X(n)$. Absolute convergence and Gamma decay justify exchanging the series and integral. Shift to $\operatorname{Re} w = 1/2 - \beta$. The apparent Gamma pole at zero is canceled by $\zeta(\rho) = 0$; its cleared product is analytic there. The remaining pole is $w = 1 - \rho$, from zeta, with residue $M_X(1)Y^{1-\rho}\Gamma(1-\rho)$. No other Gamma pole lies between these lines. The finite rectangle identities prove this residue statement and the integrability of both lines and vanishing of the horizontal edges. Taking the limit gives (5.3), since $\rho + 1/2 - \beta + \mathrm{i}u = 1/2 + \mathrm{i}(\gamma + u)$. Finiteness follows from the exponential factor: for fixed real X and $Y > 0$, the summands are dominated by $d(n)n^{-\beta}e^{-n/Y}$, whose sum is finite. The right-contour Dirichlet series is separately absolutely convergent because $\operatorname{Re}(\rho + w) > 1$. □

Uniform large tail.

Lemma 5.3. *For the choices (5.1), $0 \leq \delta \leq 1/8$, eventually in T, simultaneously at every zero with $\sigma \leq \beta < 1$ and $T \leq |\gamma| \leq 2T$,*

$$|\mathcal{T}(\rho; X, Y)| \geq 1/2.$$

Proof. Use Proposition 3.1 with any fixed $0 < \theta < 1/48$ in the integral term of (5.3). The mollifier bound is $2\sqrt{X}$. The shifted-height power is bounded by $3T^{1/6+\theta}(1 + |u|)$, and the Gamma strip has a uniform integrable $(1 + |u|)|\Gamma(1/2 - \beta + \mathrm{i}u)|$ majorant. Hence the integral term is at most a fixed multiple of

$$\sqrt{X}\,Y^{1/2-\beta}T^{1/6+\theta} = T^{\delta/2 + c(1/2-\beta) + 1/6 + \theta} \leq T^{-1/48+\theta}.$$

For the last inequality, $c(\beta - 1/2) \geq 1/4$ follows from $\beta \geq \sigma > 3/4$, while $\delta/2 \leq 1/16$. The residue term is bounded by $72T^3e^{-T}$ using $X, Y \leq T$ and Gamma exponential decay at $1 - \rho$. Thus $\mathcal{D} \to 0$ uniformly over these zeros. Since $\mathcal{D} = e^{-1/Y} + \mathcal{T}$ and $Y \to \infty$, choose the height threshold so that $e^{-1/Y} \geq 3/4$ and $|\mathcal{D}| \leq 1/4$. The reverse triangle inequality gives $|\mathcal{T}| \geq 1/2$. □

5.1 Smooth dyadic decomposition

Let S_0 be a fixed smooth nondecreasing transition function: it is zero on $(-\infty, 0]$, one on $[1, \infty)$, and takes values in $[0, 1]$. Define

$$h(x) = 1 - S_0(4x - 5), \qquad \psi(x) = h(x) - h(2x).$$

The function h is antitone, equal to one for $x \leq 5/4$ and zero for $x \geq 3/2$. Thus $0 \leq \psi \leq 1$, ψ is smooth, and its support lies in $[5/8, 3/2] \subset (1/2, 2)$. For $x > 0$,

$$\sum_{j \in \mathbb{Z}} \psi(x/2^j) = 1.$$

This is the telescoping sum of $h(x/2^j) - h(x/2^{j-1})$; the two limit values are one and zero. For a fixed x, only finitely many terms are nonzero. Set

$$B_j(\rho) = \sum_{n\geq 2} b_X(n) n^{-\rho} e^{-n/Y} \psi(n/2^j).$$

Absolute summability of the double array follows from nonnegativity of ψ and its partition identity; therefore $\sum_j B_j(\rho) = \mathcal{T}(\rho; X, Y)$ and $\sum_j |B_j(\rho)| < \infty$. Small blocks vanish for $2^j \leq X/2$ by Möbius cancellation and support; blocks with $j \leq 0$ also vanish because the $n = 1$ term was removed.

Let $R = Y(\log T)^2$ and

$$\mathcal{J}_R = \{j \in \mathbb{Z} : 0 \leq j \leq \lceil \log R/\log 2 \rceil,\ 2^j \leq R\}.$$

For T sufficiently large, $\#\mathcal{J}_R \leq 6\log T$. The discarded block norms, not merely the norm of their sum, satisfy

$$\sum_{2^j > R} |B_j(\rho)| \leq 2T^{-2}.$$

We derive this from the tail beginning at $K = \lfloor R/2 \rfloor_+$:

$$\sum_{n\geq K} |b_X(n) n^{-\rho} e^{-n/Y}| \leq \lfloor X \rfloor_+ e^{-K/Y} (1 - e^{-1/Y})^{-1}.$$

Here $|b_X(n)| \leq \lfloor X \rfloor_+$ is used, $K \geq R/4$ for $R \geq 4$, $e^{-K/Y} \leq T^{-4}$ once $\log T \geq 16$, and $(1 - e^{-1/Y})^{-1} \leq 2Y \leq 2T$. These give $2T^{-2}$. Since $|\mathcal{T}| \geq 1/2$, at least $1/4$ of the norm sum remains in $\mathcal{J}_R$; a finite pigeonhole argument gives a $j \in \mathcal{J}_R$ with

$$X/2 < 2^j \leq Y(\log T)^2, \qquad |B_j(\rho)| \geq 1/(24\log T). \tag{5.4}$$

Both the list and its cardinality are fixed before the zero varies.

6 The short-block count

6.1 Removing the real-part dependence

For a detected scale 2^j, put

$$a_n^{(j)} = b_X(n) e^{-n/Y} \psi(n/2^j), \qquad M_{j,e} = 2^{\max(\max(j,0)-1,0)+e} \quad (e = 0, 1),$$

where the exponent uses natural truncated subtraction as in Lean. A nonzero block necessarily has $j > 0$, in which case $M_{j,0} = 2^{j-1}$ and $M_{j,1} = 2^j$. The smooth block splits exactly into the two natural supports $(M_{j,0}, 2M_{j,0}]$ and $(M_{j,1}, 2M_{j,1}]$. Both halves retain the *original* smooth scale 2^j in $a_n^{(j)}$. At least one half has norm at least half of $|B_j|$.

For a general positive natural scale M and coefficients a_n with $|a_n| \leq d(n)$ on $(M, 2M]$, define

$$P_r(t) = \sum_{M<n\leq 2M} a_n (2n/M - 3)^r e^{-it\log(n/M)}, \qquad q_r(\beta) = \left(\prod_{l=0}^{r-1} \frac{\beta + l}{1 + l}\right) (-1/3)^r.$$

When $0 \leq \beta \leq 1$, $|q_r(\beta)| \leq 3^{-r}$ and $\sum_r |q_r(\beta)| \leq 3/2$. The generalized binomial series gives the exact identity

$$\sum_{r\geq 0} q_r(\beta) P_r(t) = (3/2)^\beta \sum_{M<n\leq 2M} a_n (n/M)^{-\beta} e^{-it\log(n/M)}.$$

Indeed with $z = 2n/M - 3 \in [-1, 1]$, $(2n/(3M))^{-\beta}$ has the displayed binomial expansion in $z/3$. The finite n sum and absolutely convergent r sum may be interchanged. Rescaling $n^{-\rho} = M^{-\rho}(n/M)^{-\rho}$ shows that the norm of the right-hand normalized block is $(3/2)^{\beta} M^{\beta} |\sum a_n n^{-\rho}|$. The coefficients of P_r depend on a, M, r, not on ρ.

The triangle bound $|P_r(t)| \le 2M^2$ and the geometric weights imply

$$\left| \sum_{r \ge J} q_r(\beta) P_r(t) \right| \le 3M^2 3^{-J}.$$

Take $J_T = \lceil 10 \log T \rceil$. If $M \le T^2$, the tail beginning at $J_T + 1$ is at most $3T^{-1}$ because $3^{-J_T} \le T^{-5}$. For a half-block of norm at least $1/(48 \log T)$, this tail is eventually small enough that a finite Taylor component has norm at least

$$|P_r(\gamma)| \ge \frac{M^{\beta}}{192 \log T} \ge \frac{M^{\sigma}}{192 \log T}, \qquad 0 \le r \le J_T.$$

The finite extraction does *not* cost J_T: the sum of absolute scalar weights is at most $3/2$, so some component has at least half the norm of the finite weighted sum. The scale/half/Taylor index set has cardinality at most $144(\log T)^2$.

6.2 Power-raising and convolution supports

If (5.4) has $2^j > Y^{1/2}$, it is assigned to the long alternative. Otherwise the selected half-scale satisfies

$$T^{\delta}/4 \le M \le T^{c/2} = T^{3a/4}.$$

For all sufficiently large T, there is an integer $1 \le k \le K = \lceil 2a/\delta \rceil$ with

$$T^a \le M^k \le T^{3a/2} = T^c. \tag{6.1}$$

To prove this, absorb the factor $1/4$ into $T^{\delta/2}$, set $x = \log M / \log T$ and choose $k = \lceil a/x \rceil$. If $x \le a/2$, then $kx < a + x \le 3a/2$. If $a/2 < x \le 3a/4$, then $k \le 2$ and $kx \le 3a/2$. The lower bound $kx \ge a$ and the bound $k \le \lceil 2a/\delta \rceil$ follow from the ceiling inequalities. Thus the power bound is fixed before every short scale and every zero.

For arithmetic functions supported on positive integers, Dirichlet convolution is

$$(A * B)(n) = \sum_{de=n} A(d) B(e).$$

Write A^{*k} for its k-fold iteration. Restrict the Taylor coefficients to $(M, 2M]$ by setting $A_{j,e,r}(n) = a_n^{(j)} (2n/M_{j,e} - 3)^r$ on this interval and zero elsewhere. Take the Dirichlet convolution power $A_{j,e,r}^{*k}$, not a pointwise power of coefficients. The product Dirichlet series is the kth power of the finite block series, with support in $(M^k, (2M)^k]$. We prove this finite convolution identity and its support before using it. Divide that support into the k dyadic intervals $(2^l M^k, 2^{l+1} M^k]$, $0 \le l < k$. At least one piece has norm at least $|P_r(\gamma)|^k / k$. Use the coefficientwise complex conjugate of the convolution power to change the negative phase convention to the positive detecting polynomial while preserving its norm at the *same* ordinate γ. Define

$$L_{j,e,r,k,l} = 2^l M_{j,e}^k, \qquad a_{j,e,r,k,l}(n) = T^{-\eta} \overline{A_{j,e,r}^{*k}(n)}.$$

The bounded-convolution divisor estimate makes these coefficients uniformly at most one whenever their entire support is below T^2, for fixed K and arbitrary positive η. Equations (6.1) and $c < 1$ ensure that support condition eventually, since $(2M)^k \le 2^K T^c \le T$. They

preserve the natural half-open coefficient supports. For an arithmetic function with $|A(n)| \le d(n)$, we first prove $|A^{*k}(n)| \le d(n)^{2k}$. Use the divisor subpower estimate with loss $\eta/(2(K+1))$ and $k \le K$ to obtain $|A^{*k}(n)| \le C^{2(K+1)} n^{\eta}$. To normalize on $n \le T^2$, use that estimate with loss $\eta/4$ and then absorb its fixed constant into $T^{\eta/2}$; the resulting bound is T^{η} before any family member is selected. The height loss is at most $C_K(\log T)^K$, with $C_K = 2^K(K+1)192^K$; it is eventually at most T^{η}. After normalization the selected piece therefore satisfies

$$|D_{L_q,a_q}(\gamma)| \ge L_q^{\sigma} T^{-2\eta}, \qquad T^a \le L_q \le 2^K T^c. \tag{6.2}$$

The *fixed family* consists of all tuples $q = (j, e, r, k, l)$ with $j \in \mathcal{J}_{Y(\log T)^2}$, $e = 0, 1$, $0 \le r \le J_T$, $1 \le k \le K$, $0 \le l < k$, and the two conditions in (6.1) for that half-scale. These restrictions depend on σ, δ, T, not on any zero. Its cardinality is at most $144K^2(\log T)^2$. Only the chosen index depends on the zero; the family coefficients never do. Every slab zero has either the long block or a member of this family satisfying (6.2) at its original ordinate.

6.3 Local multiplicities and short zeros

Uniform local multiplicity.

Lemma 6.1. *There is $C_b > 0$ such that for every $\sigma \ge 3/4$, $T \ge 2$ and integer k,*

$$\sum_{\substack{\rho \in \mathcal{Z}(\sigma;T,2T) \\ k \le \operatorname{Im}\rho < k+1}} m(\rho) \le C_b \log(2T).$$

Proof. For $T \ge 8$ and a nonempty bin, its center is $z_k = 2 + \mathrm{i}(k + 1/2)$; the bin location implies $T - 1 \le k \le 2T$. Each relevant zero has $\operatorname{Re}\rho \ge 7/10$ and lies inside radius $3/2$ about z_k, because its horizontal difference is at most $13/10$ and its vertical difference at most $1/2$. The disk of radius $7/4$ avoids 1. The Abel zeta growth estimate on its sphere gives $|\zeta(s)| \le 100T^3$. On the center line $\operatorname{Re} s = 2$, the absolutely convergent Möbius inverse series has norm less than $5/3$, since $\zeta(2) = \pi^2/6 < 5/3$. Its product with zeta is one, so $|\zeta(z_k)| \ge 3/5 = 0.6$. Jensen's formula for radii $3/2, 7/4$, and the finite-sum analytic-order to-divisor bridge, give

$$\sum_{\text{bin}} m(\rho) \le \frac{\log(100T^3/0.6)}{\log((7/4)/(3/2))} \ll \log(2T).$$

The bridge is necessary: it identifies each finite natural vanishing order with the nonnegative analytic divisor, rather than counting distinct roots. For $2 \le T \le T_0$, all bin sums are bounded by the finite count $N(3/4, 2T_0)$. Since $\log(2T) \ge \log 4 > 0$, this bounded contribution is absorbed by increasing the same C_b. □

For any finite subset S of these zeros, select the parity class with at least half the occupied unit bins, then one zero in each selected bin. Distinct selected bins of the same parity differ by at least two; their ordinates are one-separated. The ordinate map on the selected representatives is injective. Lemma 6.1 implies

$$\sum_{\rho \in S} m(\rho) \le 2C_b \log(2T)\, |U|, \tag{6.3}$$

where U is their ordinate set. This is a weighted finite extraction, retaining all the lost multiplicities in the factor $C_b \log(2T)$.

Let S_{short} be exactly those zeros in $\mathcal{Z}(\sigma;T,2T)$ detected by *some* fixed-family member at level $V_q = L_q^{\sigma}T^{-2\eta}$; set $N_{\mathrm{short}} = \sum_{\rho\in S_{\mathrm{short}}} m(\rho)$. Choose $\eta > 0$ with $8\eta < \sigma - 3/4$ and $\kappa = (\sigma - 3/4)/2 > 0$. Use the comparison ambient height

$$H_T = 32T(\log T)^2,$$

which eventually satisfies $T \le H_T \le T^2$ and $H_T^{1/3} \le \sqrt{T}$. The fixed scales obey $L_q \ge T^a \ge \sqrt{T}$, $L_q \le T \le H_T$ and

$$V_q = L_q^{\sigma}T^{-2\eta} \ge L_q^{\sigma-4\eta} \ge L_q^{3/4+\kappa}.$$

Thus each family member satisfies *all* hypotheses of Theorem 4.2 with height H_T; the slab is contained in an interval of that length. Partition the separated ordinate representatives into the superlevel subsets for each family member and use the finite union-cardinality bound. Combining with (6.3) gives

$$N_{\mathrm{short}} \le C\log(2T)H_T^{\eta}\sum_q \left(\frac{L_q^2}{V_q^2} + \frac{L_q^3\sqrt{H_T}}{V_q^4}\right).$$

Overlapping detection sets only enlarge this upper bound.

The two summands are, exactly,

$$L_q^{2-2\sigma}T^{4\eta}, \qquad L_q^{3-4\sigma}\sqrt{H_T}\,T^{8\eta}.$$

The first length exponent is nonnegative; use the upper length $2^K T^c$. The second is negative; use the lower length T^a. Equations (5.2), and $\sqrt{H_T} \le 32T^{1/2}\log T$, give a bound by $(2^K + 32)\log T\, T^{p+8\eta}$ per member. The family contributes at most $144K^2\log^2 T$; the bin factor at most $2\log T$; $H_T^{\eta} \le T^{2\eta}$; and the remaining four logarithms are eventually at most $T^{2\eta}$. Therefore

$$N_{\mathrm{short}} \le C_{\sigma,\delta,\eta}T^{p+12\eta}. \tag{6.4}$$

This accounts for 8η in the quotient terms, 2η in the ambient height and 2η in the logarithms. The fixed 2^K enlargement is not omitted, and no long-zero estimate was used.

7 The long-block count and the density theorem

Define the complementary zero set $S_{\mathrm{long}} = \mathcal{Z}(\sigma;T,2T) \setminus S_{\mathrm{short}}$ and its multiplicity count N_{long}. This is an actual complement, not an independently selected family of difficult zeros. Hence

$$N_{\mathrm{slab}}(\sigma;T,2T) = N_{\mathrm{short}} + N_{\mathrm{long}}. \tag{7.1}$$

The fixed-family cover implies that every member of S_{long} has a large smooth block with $2^j > Y^{1/2}$.

7.1 Uniform Mellin inversion for long blocks

For $q \ge 0$ put $\psi_q(x) = \psi(x)e^{-qx}$ and $\widehat{\psi}_q(s) = \int_0^{\infty}\psi_q(x)x^{s-1}\,\mathrm{d}x$. The compact support lies in $[5/8, 3/2]$, so this Mellin transform is entire. Its rapid vertical decay is uniform in *all* $q \ge 0$ on any fixed real strip. For each derivative order, the inequality $q^n e^{-qx} \le n!/(5/8)^n$ on the support gives uniform derivative bounds; Leibniz differentiation and Fourier decay of the logarithmic lift give $|\widehat{\psi}_q(u+\mathrm{i}v)| \le C_A(1+|v|)^{-A}$ and uniformly weighted L^1 bounds. Thus the damping parameter $q = 2^j/Y$ need not be bounded.

For $N > 0$ the smooth mollifier block is

$$B(\rho;X,N,q) = \sum_{n\ge 1} b_X(n)n^{-\rho}\psi(n/N)e^{-qn/N}.$$

It equals the normalized integral of $N^s\widehat{\psi}_q(s)\zeta(\rho+s)M_X(\rho+s)$ on $\operatorname{Re} s = 1/2$. Shift to $\operatorname{Re} s = 1/2-\beta$. The only pole is $s = 1-\rho$ and its residue is

$$R_B = N^{1-\rho}\widehat{\psi}_q(1-\rho)M_X(1).$$

There is no Gamma pole at zero in this smooth Mellin kernel; in particular the block-contour identity does not require the zero hypothesis. The smooth Mellin bounds and Abel zeta growth prove both-line integrability and vanishing horizontal edges. For $T \le |\gamma|$, $0 \le X \le T$, $1 \le N \le T^2$, the residue is at most T^{-100} eventually, uniformly in $\beta \in [3/4, 1)$ and $q \ge 0$; a sufficiently high fixed Mellin-decay power absorbs all powers of X, N, T. For $N = 2^j \ge 2$ the removed $n = 1$ term has zero smooth weight, so this is exactly $B_j(\rho)$ with $q = N/Y$.

Define, for integer $A \ge 3$,

$$w_A(u) = (1+|u|)^{-A}, \qquad \mathcal{M}_A(\gamma) = \int_{\mathbb{R}} w_A(u)Z(\gamma+u)\,\mathrm{d}u.$$

The uniform left-line Mellin bound and $|M_X(s(\gamma+u))| \le 2\sqrt{X}$ give

$$|B(\rho; X, N, q)| \le |R_B| + C\sqrt{X}N^{1/2-\beta}\mathcal{M}_A(\gamma).$$

The residue is at most $1/(48\log T)$ eventually. Hence a block of norm at least $1/(24\log T)$ satisfies $1/(48\log T) \le C\sqrt{X}N^{1/2-\beta}\mathcal{M}_A(\gamma)$. For $N > Y^{1/2}$ and $\beta \ge \sigma$, the negative exponent gives $N^{1/2-\beta} \le Y^{-(\sigma-1/2)/2}$, so every long zero satisfies

$$Y^{(\sigma-1/2)/2} \le 48C\sqrt{X}\log T\,\mathcal{M}_{128}(\gamma). \tag{7.2}$$

We have selected the fixed decay order $A = 128$ at this point.

7.2 Weighted sampling and the tail outside the height interval

For $A \ge 2$, $0 < \int w_A \le \pi$ and a nonnegative integrable weighted function f satisfies

$$\left(\int w_A f\right)^{12} \le \left(\int w_A\right)^{11}\int w_A f^{12}.$$

We prove this by the tangent inequality $12a^{11}x \le x^{12} + 11a^{12}$ for $a, x \ge 0$: take $a = (\int w_A f)/(\int w_A)$, multiply by w_A, integrate and rearrange. All three required weighted integrals are proved integrable first. For $A = 128$, coarse growth supplies the twelfth weighted integral. For a one-separated finite U the kernel row sum $\sum_{u\in U} w_{128}(t-u) \le S$ is uniform, by comparison with $(1+|t-u|)^{-2}$ and the earlier separation bound.

On the complement of $[0, 3T]$, for $u \in [T, 2T]$, $|t-u| \ge T$ and $|t| \le 3|t-u|$. Thus

$$w_{128}(t-u)Z(t)^{12} \le 12^{12}T^{-114}w_2(t-u).$$

To verify it, coarse growth gives $Z(t) \le 12(1+|t-u|)$; cancel twelve of the 128 denominator powers, reserve two powers for integrability, and use the remaining 114 powers to produce T^{-114}. Integrating gives at most $12^{12}\pi T^{-114}$ per sample. One-separated $U \subset [T, 2T]$ has $|U| \le T+1 \le 2T$, so the total sample tail costs at most $2\cdot 12^{12}\pi T^{-113}$. Consequently

$$\sum_{u\in U}\mathcal{M}_{128}(u)^{12} \le \pi^{11}S\int_{[0,3T]} Z(t)^{12}\,\mathrm{d}t + 2\cdot 12^{12}\pi^{12}T^{-113}. \tag{7.3}$$

The whole-line means have not been replaced by a compact integral without accounting for the tails.

Select weighted separated representatives of S_{long} using (6.3), preserving the property (7.2). Raise (7.2) to the twelfth power and sum over their ordinate set:

$$|U|Y^{6\sigma-3} \le (48C)^{12}X^6(\log T)^{12}\sum_{u\in U}\mathcal{M}_{128}(u)^{12}.$$

Here $(\sqrt{X})^{12}=X^6$ and $(Y^{(\sigma-1/2)/2})^{12}=Y^{6\sigma-3}$ exactly. Use (7.3) and the local multiplicity factor $\log(2T)\le 2\log T$. The result is the unconditional reduction

$$N_{\mathrm{long}}Y^{6\sigma-3} \le CX^6(\log T)^{13}\left(\int_{[0,3T]} Z(t)^{12}\,\mathrm{d}t + T^{-113}\right). \tag{7.4}$$

Its right side contains the actual moment, not an assumed upper bound on it.

7.3 The exponent saving and the final parameter choice

Use Theorem 1.3 with a positive loss $\nu \le (1-c)/8$. If its constant is $D>0$, then for $T\ge 1$ eventually,

$$\int_{[0,3T]} Z^{12} + T^{-113} \le (D+1)T^{2+\nu}.$$

Dividing (7.4) by the positive $Y^{6\sigma-3}$ gives the scalar factor

$$\frac{X^6(\log T)^{13}T^{2+\nu}}{Y^{6\sigma-3}} = (\log T)^{13}T^{p-(1-c)+6\delta+\nu}.$$

Since $6\delta < (1-c)/2$, the pre-logarithmic exponent is less than $p-3(1-c)/8$. Absorb $(\log T)^{13}$ using the further allowance $(1-c)/8$ to obtain

$$N_{\mathrm{long}} \le C'T^{p-(1-c)/4} \tag{7.5}$$

eventually. This fixed saving is possible precisely because $c<1$.
We now prove Theorem 1.2.

Proof. First suppose $3/4<\sigma<1$. Given $\varepsilon>0$, choose, in the same order as the argument,

$$\delta=\min(1/16,(1-c)/24),\qquad \eta=\min(\varepsilon/24,(\sigma-3/4)/16),\qquad \nu=(1-c)/16.$$

They satisfy $\delta>0$, $\delta\le 1/8$, $\delta<(1-c)/12$, $\eta>0$, $8\eta<\sigma-3/4$, $12\eta\le\varepsilon$, and $0<\nu\le(1-c)/8$. Choose the actual moment constant and threshold with this ν. Thus every parameter, power bound and coefficient family is fixed before heights and zeros vary.

Combine the exact partition (7.1) with (6.4) and (7.5). Since $T\ge 1$ and $1-c>0$, the long power is at most $T^{p+12\eta}$. Therefore

$$N_{\mathrm{slab}}(\sigma;T,2T)\le C_sT^r,\qquad r=p+12\eta>0,$$

eventually in T.

We detail the conversion from eventual slabs to a bound at every height, because dropping the compact part would weaken the stated endpoint. Let $R\ge 1$ exceed the slab threshold and $Q=N_{\mathrm{slab}}(\sigma;0,2R)$. If $1\le T<R$, monotonicity gives $N_{\mathrm{slab}}(\sigma;T,2T)\le Q\le QT^r$; if $T\ge R$, use the eventual bound. Hence the same estimate, with constant C_s+Q, holds for every $T\ge 1$. Put $b=2^r>1$, $Q_0=N_{\mathrm{slab}}(\sigma;0,1)$ and $A=Q_0+(C_s+Q)/(b-1)>0$. Then $A+(C_s+Q)\le Ab$. Inductively split the inclusive interval $[0,2^{k+1}]$ at 2^k to obtain

$$N_{\mathrm{slab}}(\sigma;0,2^k)\le A(2^k)^r.$$

The shared boundary may be counted twice by this inequality, which is harmless for an upper bound and preserves every zero. For arbitrary $T \geq 1$, choose $2^k \leq T < 2^{k+1} \leq 2T$. Monotonicity yields $N_{\mathrm{slab}}(\sigma; 0, T) \leq AbT^r$. Conjugation and its multiplicity identity give $N(\sigma, T) \leq 2AbT^r$. Since $r = p + 12\eta \leq p + \varepsilon$ and $T \geq 2$, enlarge the power to $T^{p+\varepsilon}$. The resulting constant is positive and chosen before all T.

For $\sigma \geq 1$ the zero set is empty by Section 2. Take $C = 1$; the right side is positive for $T \geq 2$, even if its exponent is negative. Finally use the exact membership lemma to identify $\mathcal{Z}(\sigma; -T, T)$ with the finite set of *all* zeros in the registered statement. Lemma 2.2 supplies positive finite multiplicity and the equality of natural and extended order at every member. The natural-to-real sum casts merely change the numerical encoding of the same sum. This proves the complete endpoint, including its witnesses and all boundary conventions. □

A A contour proof of the fourth-moment bound

The fourth moment is classical; the mean-value theorem of Ingham [6] already gives the bound required here. We retain this contour proof because it explains the squared-zeta representation used again in Appendix C. It constructs a one-sided contour integral, truncates an ordinary-divisor polynomial, and applies a mean-square inequality. The latter is the standard Dirichlet-polynomial mean-value argument (cf. [9]), with its constants retained.

A.1 A one-sided contour representation

Let $\Lambda_0(s) = \Lambda(s) + s^{-1} + (1-s)^{-1}$ away from $0, 1$, and take its entire extension at those two points. Thus

$$\Lambda(s) = \Lambda_0(s) - s^{-1} - (1-s)^{-1}.$$

Define the entire function

$$\Xi_0(s) = s(1-s)\Lambda_0(s) - 1.$$

For $s \neq 0, 1$, it equals $s(1-s)\Lambda(s)$ and satisfies $\Xi_0(1-s) = \Xi_0(s)$. In particular it is the pole-cleared numerator, not a new independently specified zeta function. Put

$$H_*(w) = (1 - 4w^2)^4, \qquad A_t = (s(t)(1 - s(t)))^2 = (1/4 + t^2)^2,$$
$$F_t(w) = \frac{e^{100w^2} H_*(w) \Xi_0(s(t) + w)^2}{A_t}, \qquad \mathscr{I}_t(w) = F_t(w)/w.$$

The denominator A_t never vanishes. The numerator F_t is entire, $F_t(0) = \Lambda(s(t))^2$, and $F_t(-w) = F_{-t}(w)$.

With $t \in \mathbb{R}$ and $c > 0$, set

$$J_c(t) = \frac{1}{2\pi} \int_{\mathbb{R}} \mathscr{I}_t(c + \mathrm{i}u)\, \mathrm{d}u, \quad B_t = \Gamma_{\mathbb{R}}(s(t))\Gamma_{\mathbb{R}}(s(-t)), \quad R(t) = J_1(-t)/B_t.$$

The following identity reduces the fourth moment to this one-sided integral.

A one-sided squared-zeta representation.

Lemma A.1. *For every $t \in \mathbb{R}$ and $c > 0$, the integrals defining $J_c(t)$ converge absolutely, $B_t \neq 0$, and*

$$\Lambda(s(t))^2 = J_c(t) + J_c(-t), \qquad Z(t)^2 = 2 \operatorname{Re} R(t).$$

Proof. Integrate $\mathscr{I}_t$ around the rectangle with vertical edges $-c, c$ and horizontal heights $\pm U$. The only pole is the simple pole at 0 and its residue is $F_t(0)$. Reflection $w \mapsto -w$ converts the left vertical edge to the right vertical edge for $-t$. We prove a polynomial bound for

Ξ_0 on each fixed vertical strip and a Gaussian-polynomial majorant for $F_t(x+\mathrm{i}u)$, uniform for $|x| \le c$. It follows that the horizontal edges tend to zero and both vertical integrals are absolutely integrable. Taking the limit of the finite rectangle identity gives the first formula, with its normalization $1/(2\pi)$ (equivalently $1/(2\pi\mathrm{i})$ against $\mathrm{d}w$).

On the critical line the functional equation gives $\Lambda(s(t)) = \Lambda(s(-t))$, whereas conjugation gives $\Lambda(s(-t)) = \overline{\Lambda(s(t))}$. Thus the completed critical-line value is real. The Gamma factors are nonzero, and

$$\Lambda(s(t))^2 = Z(t)^2 B_t.$$

Conjugation of the right-hand integrands gives $J_1(t) = \overline{J_1(-t)}$ and $B_t \in \mathbb{R}$. $\Lambda(s(t))^2 = J_c(t) + J_c(-t)$, divided by B_t, supplies $Z(t)^2 = 2\,\mathrm{Re}\,R(t)$. Squaring and using $|\mathrm{Re}\,z| \le |z|$ also gives $Z(t)^4 \le 4|R(t)|^2$, which is the inequality needed for the fourth moment. □

The arithmetic identity used to open the right contour is

$$\zeta(s)^2 = \sum_{n\ge1} d(n) n^{-s}, \qquad \mathrm{Re}\,s > 1.$$

It is proved locally from the Dirichlet convolution of the constant-one arithmetic function with itself, which counts divisors, and multiplication of absolutely convergent Dirichlet series. It is not inferred at $\mathrm{Re}\,s = 1/2$. For a shift w, put

$$P_t(w) = \left(\frac{(s(-t)+w)(1-s(-t)-w)}{s(-t)(1-s(-t))}\right)^2,$$
$$\mathcal{K}_t(w) = \frac{e^{100w^2} H_*(w)}{w} P_t(w) \frac{\Gamma_{\mathbb{R}}(s(-t)+w)^2}{B_t}.$$

On $\mathrm{Re}\,w = 1$ the squared-zeta series is absolutely convergent and the exact normalized integral is

$$R(t) = \frac{1}{2\pi} \int_{\mathbb{R}} \mathcal{K}_t(1+\mathrm{i}u) \sum_{n\ge1} d(n) n^{-s(-t)-1-\mathrm{i}u}\,\mathrm{d}u.$$

We prove summability of the integrals of term norms before interchanging the sum and this integral.

A.2 Gamma shifts, finite prefixes and the tail

The kernel estimates are derived, not supplied as an asymptotic shortcut. The local digamma development starts from Euler's Gamma limit and the convergent series

$$\psi_\Gamma(z) + \gamma_E = \sum_{m\ge0} \left(\frac{1}{m+1} - \frac{1}{z+m}\right), \qquad \mathrm{Re}\,z > 0,$$

where $\psi_\Gamma = \Gamma'/\Gamma$ and γ_E is Euler's constant. Comparing the reciprocal sum with the telescoping logarithmic integral gives

$$|\psi_\Gamma(z) - \log z| \le 4/|\,\mathrm{Im}\,z| \qquad (|\,\mathrm{Im}\,z| \ge 1,\ \mathrm{Re}\,z > 0).$$

The normalized shift

$$\left(\frac{\Gamma_{\mathbb{R}}(s(-t)+vw)}{\Gamma_{\mathbb{R}}(s(-t))}\right)^2 e^{-vw\ell_t}, \qquad \ell_t = \log(t/(2\pi)) - \mathrm{i}\pi/2,$$

has initial value one and logarithmic derivative bounded by $(17|w| + 2|w|^2)/t$ when $t \ge 4$, $\mathrm{Re}\,w \ge 0$, $|w| \le t/2$, $0 \le v \le 1$. Integrating the derivative bound gives both its exponential bound and its difference-from-one bound. The pole factor $P_t(w)$ is bounded and has

difference from one at most $12|w|/t$ on this range. For the far $|u|$ range, reciprocal Gamma reflection and elementary exponential estimates are multiplied by e^{-100u^2}. These give, for every fixed $c > 0$, a constant $K_c > 0$ such that

$$|\mathcal{K}_t(c + \mathrm{i}u)| \le K_c t^c e^{-98u^2} \qquad (t \ge 4,\ t \ge 4c,\ u \in \mathbb{R}). \tag{A.1}$$

The estimates explicitly separate near and far shifts and prove their integrability.

For $c > 0$ and finite $S \subset \mathbb{N}$, or $S = \{n \in \mathbb{N} : n > N\}$ on a line $c > 3/2$, define

$$R_S(t; c) = \frac{1}{2\pi} \int_{\mathbb{R}} \mathcal{K}_t(c + \mathrm{i}u) \sum_{n \in S} d(n) n^{-1/2-c+\mathrm{i}(t-u)} \, \mathrm{d}u.$$

The $n = 0$ term is defined to be zero. Each summand has no pole between two positive shift lines; the Gamma arguments remain in the positive half-plane, the explicit $1/w$ pole stays to the left, and the Gaussian kills horizontal edges. Thus its integral is independent of the chosen positive line. Shift the finite prefix to a small line c, and shift its complement to a large line $b > 3/2$. The exact identity is

$$R(t) = R_{\{0,\dots,N\}}(t; c) + R_{\{n>N\}}(t; b).$$

This is an exact decomposition, not an approximate functional equation assumed for the prefix. The complement is absolutely convergent on $\operatorname{Re} w = b$. Since $d(n) \le n$, comparison with an integral gives

$$\sum_{n>N} d(n) n^{-1/2-b} \le \frac{N^{3/2-b}}{b - 3/2}.$$

From (A.1),

$$|R_{\{n>N\}}(t; b)| \le K_b' t^b N^{3/2-b}.$$

If $N \ge H^{1+q}$, $H \le t \le 2H$ and $q > 0$, choose $b = 2 + 2/q$. The identity

$$b + (1 + q)(3/2 - b) = -(1 + q)/2$$

then bounds the entire tail by a constant, uniformly in H, t, N and in the positive prefix shift c.

A.3 Mean squares of Dirichlet polynomials

Dyadic Dirichlet mean square.

Lemma A.2. *For arbitrary complex coefficients a_n, positive integer N and $H \ge 0$,*

$$\int_0^H \left| \sum_{N<n\le 2N} a_n n^{-\mathrm{i}t} \right|^2 \mathrm{d}t \le (H + 2(5\pi + 1)N) \sum_{N<n\le 2N} |a_n|^2.$$

The same bound is invariant under translation or reflection of the integration variable, with the interval length replacing H.

Proof. Write

$$Q_N(a) = \sum_{\substack{N<m,n\le 2N \\ m\ne n}} \frac{\overline{a_n} a_m}{\log m - \log n}, \qquad a_n^{(H)} = a_n n^{-\mathrm{i}H}.$$

Integration of each exponential term gives the exact identity

$$\int_0^H \left| \sum a_n n^{-\mathrm{i}t} \right|^2 \mathrm{d}t = H \sum |a_n|^2 + \mathrm{i}(Q_N(a^{(H)}) - Q_N(a)),$$

where the real left side is viewed inside $\mathbb{C}$. For $x, y > 0$, $x \neq y$, elementary bounds for $\log(y/x)$ give

$$\left| \frac{1}{\log y - \log x} - \frac{x}{y - x} \right| \leq 1.$$

Consequently Q_N is the sum of a main discrete Hilbert form and an error whose norm is at most $(\sum |a_n|)^2 \leq N \sum |a_n|^2$.

For completeness, the discrete Hilbert form here is obtained directly from Fourier orthogonality. For finite integer frequencies S and coefficient families a, b, define

$$\mathcal{H}_S(a, b) = \sum_{\substack{m,n \in S \\ m \neq n}} \frac{\overline{a_n} b_m}{m - n}, \quad A(x) = \sum_{n \in S} a_n e^{\mathrm{i}nx}, \quad B(x) = \sum_{n \in S} b_n e^{\mathrm{i}nx}.$$

Termwise integration by parts yields

$$\int_0^{2\pi} (\pi - x) \overline{A(x)} B(x) \, \mathrm{d}x = 2\pi \mathrm{i} \mathcal{H}_S(a, b).$$

Use $|\pi - x| \leq \pi$, the elementary product bound $|A||B| \leq |A|^2 + |B|^2$, and Fourier orthogonality $\int_0^{2\pi} |A|^2 = 2\pi \sum |a_n|^2$ to obtain

$$|\mathcal{H}_S(a, b)| \leq \pi \left(\sum |a_n|^2 + \sum |b_n|^2 \right).$$

Take the scaled first family $\widetilde{a}_n = (n/N) a_n$, whose square mass is at most $4 \sum |a_n|^2$. The main logarithmic form is $N \mathcal{H}_S(\widetilde{a}, a)$, so it has norm at most $5\pi N \sum |a_n|^2$. Adding the reciprocal-log error gives $|Q_N(a)| \leq (5\pi + 1) N \sum |a_n|^2$. The endpoint twist preserves square mass, so the two endpoint forms give the claimed factor 2. A translation only twists coefficients by unit phases; reflection conjugates or reverses those phases. Thus the same estimate holds on translated intervals. □

Fourth moment.

Proposition A.3. *For every $\eta > 0$ there are $C \geq 0$ and H_0 such that for $H \geq H_0$, $H > 0$,*

$$\int_H^{2H} Z(t)^4 \, \mathrm{d}t \leq C H^{1+\eta}.$$

Proof. Choose $q = \min(\eta, 1)/20$. Thus $0 < q \leq 1/20$ and $7q \leq \eta$. Choose a dyadic length $N = 2^M$ with $H^{1+q} \leq 2^M \leq 2H^{1+q}$. The preceding tail bound gives $|R(t) - R_{\{0,\ldots,N\}}(t; q)| \leq K_t$ uniformly on $[H, 2H]$. Define

$$a_n = d(n) n^{-1/2-q}, \qquad F(t) = \int_{\mathbb{R}} e^{-98u^2} \left| \sum_{1 \leq n \leq N} a_n n^{-\mathrm{i}(u-t)} \right|^2 \mathrm{d}u.$$

The finite prefix representation and (A.1), followed by weighted Cauchy–Schwarz with Gaussian mass $\sqrt{\pi/98}$, imply $|R_{\{0,\ldots,N\}}(t; q)|^2 \leq K_g t^{2q} F(t)$. Lemma A.1 and $|x+y|^2 \leq 2|x|^2 + 2|y|^2$ give

$$Z(t)^4 \leq 8 K_g (2H)^{2q} F(t) + 8 K_t^2.$$

All Gaussian-product functions are integrable: a finite polynomial is bounded by its coefficient norm sum, and the Gaussian is integrable. Fubini may therefore be applied.

The divisor subpower estimate gives, for every $q > 0$, the uniform bound

$$\sum_{L < n \leq 2L} |d(n) n^{-1/2-c}|^2 \leq D (2L)^q \qquad (c \geq 0).$$

Partition the prefix into its $n = 1$ term and dyadic blocks. Finite Cauchy–Schwarz costs $(M+1)^2$ after the common block bound is used. Lemma A.2 then gives

$$\int_H^{2H} F(t)\,\mathrm{d}t \le D_p(M+1)^2(H + 2(5\pi+1)2^M)(2^M)^q.$$

We absorb $(M+1)^2$ into a fixed multiple of $(2^M)^q$. With $2^M \le 2H^{1+q}$ the full scaled expression is bounded by $D_b H^{1+7q}$: before the final simplification its power is $1+5q+2q^2 \le 1+7q$ for $q \le 1$. Integrating the pointwise inequality gives

$$\int_H^{2H} Z(t)^4\,\mathrm{d}t \le 8K_g D_p D_b H^{1+7q} + 8K_t^2 H \le (8K_g D_p D_b + 8K_t^2)H^{1+\eta}.$$

The height threshold is chosen so that every kernel, tail and dyadic-cutoff condition holds before t varies. This is the estimate consumed later. □

B Pointwise values and local second moments

The inequality in this appendix is Heath-Brown's Lemma 3 [4], and the subsequent summation is the step behind equation (44) of that paper. The details below retain the rectangle, moving residue, and kernel convolution used in the registered reconstruction. For $t \in \mathbb{R}$ and $L \ge 0$, define

$$M(t,L) = \int_{-L}^{L} e^{-|u|} Z(t+u)^2\,\mathrm{d}u = \int_{t-L}^{t+L} e^{-|t-v|} Z(v)^2\,\mathrm{d}v.$$

The second equality is exactly translation of the first, with no kernel replacement. Also put

$$M_\infty(t) = \int_{\mathbb{R}} e^{-|u|} Z(t+u)^2\,\mathrm{d}u, \quad M_\delta^*(t) = \int_{\mathbb{R}} \frac{e^{-4|u|/3}}{\delta+|u|} Z(t+u)^2\,\mathrm{d}u \quad (\delta > 0).$$

All these integrals are genuine convergent integrals by coarse quadratic growth and exponential decay.

Pointwise control by a local mean.

Proposition B.1. *There is $C > 0$ such that, for every real $t \ge 10$,*

$$Z(t)^2 \le C \log t\,(1 + M(t, (\log t)^2)).$$

Proof. Here the argument parameter is $\delta = 1/\log t$ and the rectangle height is $R = (\log t)^2$. The initial finite rectangle integrates $\Gamma(w)\zeta(s(t)+w)^2$ over $-\delta \le \operatorname{Re} w \le \delta$, $|\operatorname{Im} w| \le R$. Since $t \ge 10$, $\delta < 1/2$, the moving zeta pole has real part $1/2$ and is strictly to the right of this rectangle. Thus only the Gamma pole at zero is enclosed; its residue is $\zeta(s(t))^2$. The uniform $\delta \le 1/4$ estimates below are used initially for sufficiently large t, and the compact remaining range is handled at the end.

To bound its vertical sides, we first establish an off-critical inequality. For $s = 1/2+\delta+\mathrm{i}t$, Mellin inversion of the actual exponentially smoothed divisor series gives

$$\zeta(s)^2 = \sum_{n\ge1} d(n) n^{-s} e^{-n} - \operatorname{Res}_{w=1-s}(\Gamma(w)\zeta(s+w)^2) - \frac{1}{2\pi}\int_{\mathbb{R}} \Gamma(-\delta+\mathrm{i}u)\zeta(1/2+\mathrm{i}(t+u))^2\,\mathrm{d}u.$$

The shift crosses the Gamma pole at zero and the moving double zeta pole $w = 1-s$. The latter is evaluated by differentiating its cleared numerator: if $\zeta_1(z) = z\zeta(1+z)$ is the analytic pole-removed function with $\zeta_1(0) = 1$, its residue is

$$\Gamma(1-s)\psi_\Gamma(1-s) + 2\Gamma(1-s)\zeta_1'(0).$$

The smoothed series is bounded absolutely by $\sum ne^{-n}$; the residue is uniformly bounded for $0 < \delta \le 1/4$, $t \ge 10$ using Gamma decay and a digamma bound. All horizontal limits and vertical integrability are proved with exponential or polynomial strip majorants. Hence

$$|\zeta(1/2 + \delta + \mathrm{i}t)|^2 \le C(1 + M^*_\delta(t)).$$

The functional equation transfers this to $1/2 - \delta + \mathrm{i}t$, with an additional factor at most $\exp(2\delta(\log(t/2 + 2) + D))$. This factor is obtained by integrating a Gamma logarithmic-derivative bound on the fixed positive-real-part strip. It is not assumed to be one.

The Gamma bounds on the small vertical lines are

$$(\delta + |u|)|\Gamma(\pm\delta + \mathrm{i}u)| \le 144e^{-4|u|/3}.$$

After substitution of the off-critical inequalities, the vertical-edge majorant contains a convolution of two kernels $k_\delta(u) = e^{-4|u|/3}/(\delta + |u|)$. The argument explicitly proves

$$\int_{\mathbb{R}} k_\delta(w)k_\delta(u - w)\,\mathrm{d}w \le \frac{\pi}{\delta}e^{-|u|}.$$

Indeed the excess exponential reserve bounds the product by $e^{-|u|}/2$ times the sum of the squares of $r_\delta(x) = e^{-|x|/6}/(\delta + |x|)$ at w and $u - w$. Each square is at most $(\delta^2 + x^2)^{-1}$, whose integral is π/δ. Tonelli and the proved product majorants then yield the convolution inequality and bound both vertical edges by $C\delta^{-1}(1 + M_\infty(t))$.

Eventually $R \le t/2$, so every shifted height on the edges is still large. For $|u| \le R$ the reflected Gamma factor above is at most e^4 when $D \le \log t$; this is proved before it is absorbed into the constant. The horizontal edges have bounds $Ce^{-4R/3}(5(1 + t + R))^2$ and are eventually at most one, since $R = (\log t)^2$. The residue rectangle therefore gives

$$Z(t)^2 \le C \log t(1 + M_\infty(t))$$

eventually.

Finally, if $|u| \ge (\log t)^2$ and t is sufficiently large, coarse quadratic growth gives $e^{-|u|}Z(t + u)^2 \le 19C_0e^{-|u|/2}$. Its tail integral is uniformly bounded. Thus $M_\infty(t) \le M(t, (\log t)^2) + D_0$, and the last additive constant is absorbed into $1 + M$. Continuity of Z bounds the remaining compact range $10 \le t \le t_0$; since $\log t \ge \log 10 > 0$ and $M \ge 0$, increasing C extends the estimate to every $t \ge 10$. □

Exponential overlap and the finite-peak inequality.

Lemma B.2. *For a one-separated finite W and any $x \in \mathbb{R}$, $\sum_{t\in W} e^{-|t-x|} \le 4$. Suppose $W \subset [c - G/2, c + G/2]$, $G, L \ge 0$, $G/2 + L \le G$, $c - G/2 \ge 10$, $c + G/2 \le T$, $T \ge 10$, $(\log t)^2 \le L$ and $Z(t) \ge V > 0$ for $t \in W$. Then*

$$V^2|W| \le C \log T \left(|W| + 4\int_{c-G}^{c+G} Z(u)^2\,\mathrm{d}u\right).$$

Proof. For each integer $k \ge 0$, at most two points of W have $k \le |t - x| < k + 1$, one on each side, by one-separatedness. On this shell $e^{-|t-x|} \le 2^{-k}$. Summing $2\sum_{k\ge0} 2^{-k} = 4$ proves the overlap bound. Proposition B.1 gives the displayed point inequality at each t; enlarge its integration radius from $(\log t)^2$ to L and bound $\log t \le \log T$. Every resulting interval $[t - L, t + L]$ lies in $[c - G, c + G]$ by the fit condition. Interchange the finite sum and integral, apply the overlap bound to the nonnegative integrand, and sum the constant terms. This is exactly the equation-(44) sum interface. □

C Voronoi summation and the stationary expansion

The oscillatory phase in this appendix belongs to the theory originating with Atkinson [1] and used in Heath-Brown's moment argument [4]. Here the local-mean expansion is reconstructed through an exact compact-test divisor Voronoi formula and a stationary-phase calculation. We do not replace it by an unspecified version of Atkinson's formula: the weights, both signs, and width restrictions of the present calculation matter later.

We assume $T > 0$, $G > 0$, $L > 0$, and positive summation indices in this appendix unless stated otherwise. Zero-indexed terms containing negative powers are assigned the value zero.

C.1 Freezing the Gamma factor and Gaussian averaging

Put

$$\chi_t = \frac{\Gamma_{\mathbb{R}}(s(-t))}{\Gamma_{\mathbb{R}}(s(t))}, \qquad |\chi_t| = 1, \qquad \ell_t = \log(t/(2\pi)) - \mathrm{i}\pi/2.$$

The divisor weight is the entire function

$$\Omega(q) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{e^{100(1+\mathrm{i}u)^2} H_*(1+\mathrm{i}u) e^{-q(1+\mathrm{i}u)}}{1+\mathrm{i}u} \, \mathrm{d}u.$$

It obeys $\Omega(q) + \Omega(-q) = 1$. To see this, shift the line from 1 to -1; the numerator is entire, its value at zero is one, and the only pole of its quotient by w is at zero. Gaussian horizontal decay permits the limit of the finite rectangle. The left line, after $w \mapsto -w$, is the negative of the right line for $-q$. Thus their difference is one. The same Gaussian majorants justify differentiation under the integral, so Ω is entire, with $\Omega'(q) = -(2\pi)^{-1} \int e^{100w^2} H_*(w) e^{-qw} \, \mathrm{d}u$, where $w = 1 + \mathrm{i}u$. On a fixed imaginary strip it satisfies

$$|\Omega(q)| \le C \min(1, e^{-\operatorname{Re} q}),$$
$$|\Omega(q+h) - \Omega(q)| \le C|h| e^{|h|} \min(e^{-\operatorname{Re} q}, e^{\operatorname{Re} q}) \quad (h \in \mathbb{R}).$$

The first decay follows on the right contour and then by reflection; the second follows by integrating the derivative and using the same reflection. These estimates precede all physical heights and indices.

The Gamma shift calculation from Section A approximates the normalized right integral by

$$\chi_t \sum_{n \ge 1} d(n) n^{-1/2+\mathrm{i}t} \Omega(\log n - \ell_t)$$

with a bounded error. The exact near-kernel error has an integrable majorant $Ce^{-90u^2}(1 + |u|)^{12}$, and the far-kernel part has the same form. Coefficientwise errors sum because $\sum d(n) n^{-3/2} < \infty$. Freeze the weight at a center T to define

$$F(T, x) = \chi_{T+x} \sum_{n \ge 1} d(n) n^{-1/2+\mathrm{i}(T+x)} \Omega(\log n - \ell_T).$$

For $T \ge 8$, $|x| \le T/2$, every fixed $\theta > 0$ gives

$$|Z(T+x)^2 - 2 \operatorname{Re} F(T, x)| \le C_\theta (1 + |x| T^{-1/2+\theta}). \tag{C.1}$$

Indeed $\log(T+x) - \log T$ has absolute value at most $2|x|/T$, and the weight difference bound can be summed against $d(n) n^{-1/2} \min(1, T/(2\pi n))$; this mass is $O_\theta(T^{1/2+\theta})$. This is the freezing error, not a formal replacement of the height-dependent phase by a constant.

For $G > 0$ define

$$A(T,G) = G^{-2} + \mathrm{i}/(2T), \quad Q(T,G;v) = \int_{\mathbb{R}} e^{\mathrm{i}vx} e^{-A(T,G)x^2}\,\mathrm{d}x = \left(\frac{\pi}{A(T,G)}\right)^{1/2} e^{-v^2/(4A(T,G))}.$$

The real part of A is positive; the principal half-power is therefore the one in the convergent complex Gaussian formula. If $G^2 \le 2T$, then

$$|Q(T,G;v)| \le \sqrt{\pi} G e^{-(Gv)^2/8}. \tag{C.2}$$

This follows from $|\operatorname{Im} A| \le \operatorname{Re} A$, $\operatorname{Re} A^{-1} \ge 1/(2\operatorname{Re} A)$ and the norm of the half-power.

The logarithmic derivative of χ_t is $\mathrm{i}(\log\pi - \operatorname{Re}\psi_\Gamma(s(t)/2))\chi_t$, and the quantity in parentheses is $-\log(t/(2\pi)) + O(1/t)$. On $|x| \le r \le T/2$ we prove

$$|\chi_{T+x} - \chi_T e^{-\mathrm{i}(x\log(T/(2\pi)) + x^2/(2T))}| \le (18/T + 2r^2/T^2)|x|.$$

Integrating this error on $[-r,r]$ against the physical Gaussian $e^{-(x/G)^2}$ costs at most $2r^2(18/T + 2r^2/T^2)$. The two omitted Gaussian tails cost $2\sqrt{2\pi}Ge^{-(r/G)^2/2}$. Thus the actual phase-Gaussian transform is compared to $\chi_T Q(T,G;v - \log(T/(2\pi)))$, with both errors stated and proved. Summability of frozen coefficients and product Gaussian majorants justify the integral/series exchanges.

C.2 A compact test function and its lattice phase

Recall the fixed transition S_0 from Section 5. For $L > 0$ set

$$b_v = \frac{T}{2\pi} e^{v/G}, \qquad \vartheta_{T,G,L}(x) = S_0\!\left(\frac{x - b_{-2L}}{b_{-L} - b_{-2L}}\right) S_0\!\left(\frac{b_{2L} - x}{b_{2L} - b_L}\right).$$

It equals one on $[b_{-L}, b_L]$ and is supported in $[b_{-2L}, b_{2L}]$. The compact test is

$$g_{T,G,L}(x) = \vartheta_{T,G,L}(x) x^{-1/2+\mathrm{i}T} \Omega(\log x - \ell_T)\chi_T Q(T,G;\log x - \log(T/(2\pi)))e^{-2\pi\mathrm{i}x} \quad (x > 0), \tag{C.3}$$

and is zero off the positive support interval. The phase $e^{-2\pi\mathrm{i}x}$ is not cosmetic: it is identically one on the integer lattice, so it preserves the discrete divisor sum, but it is retained in every subsequent continuous Voronoi integral. Removing it would alter the saddles. Define $\Sigma(T,G,L) = \sum_{n\ge1} d(n) g_{T,G,L}(n)$. The compact test has a positive lower support endpoint, an upper endpoint, smoothness on $\mathbb{R}$, and support contained between those endpoints. Hence its Mellin transform is entire and decreases faster than any fixed power on each vertical line.

For every fixed $\delta > 0$, eventually for $T^\delta \le G \le T^{1/2-\delta}$, the preceding Gaussian averaging argument, with $L = \log T$, yields

$$\left| \int_{\mathbb{R}} e^{-((t-T)/G)^2} Z(t)^2\,\mathrm{d}t - 2\operatorname{Re}\Sigma(T,G,\log T) \right| \le C_\delta G \log T. \tag{C.4}$$

We obtain this by integrating (C.1), replacing the actual Gamma Gaussian transform by its quadratic one, restricting to $|G(\log n - \log(T/(2\pi)))| \le \log T$, and smoothing the band. Each omitted term is bounded before summation; (C.2) makes the band complement exponentially small. The freezing and quadratic errors use fixed positive losses smaller than the width margin δ. On $[T-G, T+G]$, $e^{-((t-T)/G)^2} \ge e^{-1}$, so (C.4) also implies

$$I(T,G) := \int_{T-G}^{T+G} Z(t)^2\,\mathrm{d}t \le 2e\operatorname{Re}\Sigma(T,G,\log T) + C_\delta G\log T. \tag{C.5}$$

C.3 The divisor Voronoi formula

For clarity, the Bessel kernels used here have the integral representations

$$K_0(x) = \int_0^\infty e^{-x\cosh u}\,\mathrm{d}u,$$
$$Y_0(x) = \frac{2}{\pi}\left(\int_0^{\pi/2} \sin(x\cos u)\,\mathrm{d}u - \int_0^\infty \frac{e^{-xu}}{\sqrt{1+u^2}}\,\mathrm{d}u\right), \quad x > 0.$$

The divisor Voronoi formula.

Lemma C.1. *Let $g : \mathbb{R} \to \mathbb{C}$ be C^∞ and supported in $[a, b]$ with $0 < a \le b$. Then*

$$\begin{aligned}\sum_{n\ge1} d(n)g(n) = \int_0^\infty (\log x + 2\gamma_E)g(x)\,\mathrm{d}x \\ - 2\pi\sum_{n\ge1} d(n)\int_0^\infty g(x)Y_0(4\pi\sqrt{nx})\,\mathrm{d}x \\ + 4\sum_{n\ge1} d(n)\int_0^\infty g(x)K_0(4\pi\sqrt{nx})\,\mathrm{d}x.\end{aligned} \tag{C.6}$$

Both dual series converge absolutely.

Proof. The initial Mellin integral is $(2\pi\mathrm{i})^{-1}\int_{(2)}\zeta(s)^2\mathcal{M}g(s)\mathrm{d}s$. Termwise Mellin inversion gives the divisor sum, since the coefficient series is absolutely convergent on this line. Shift to $\operatorname{Re} s = -1/2$. The double pole at $s = 1$ has cleared numerator $((s-1)\zeta(s))^2\mathcal{M}g(s)$. Since $\zeta(s) = (s-1)^{-1} + \gamma_E + O(s-1)$, its derivative at 1 equals $(\mathcal{M}g)'(1) + 2\gamma_E\mathcal{M}g(1)$, hence the integral main term in (C.6). Smooth compact support proves arbitrary Mellin vertical decay and entireness, and coarse zeta strip growth makes the horizontal edges vanish; the infinite-line shift is therefore a limit of proved finite rectangle identities.

The actual functional equation splits the reflected squared-zeta multiplier into two exact multipliers. If $A(s) = (2\pi)^{-s}\Gamma(s)$, these are

$$M_-(z) = A(1-z)^2(e^{\pi\mathrm{i}(1-z)} + e^{-\pi\mathrm{i}(1-z)}), \qquad M_+(z) = 2A(1-z)^2.$$

For $n \ge 1$ define $\mathcal{V}_\pm[g](n)$ by

$$\mathcal{V}_\pm[g](n) = \frac{1}{2\pi\mathrm{i}}\int_{(-1/2)} n^{z-1}M_\pm(z)\mathcal{M}g(z)\mathrm{d}z.$$

Both multipliers are holomorphic on $\operatorname{Re} z < 1$; on the initial dual line their norms are bounded by a quadratic polynomial in $|\operatorname{Im} z|$. Absolute Mellin integrability, followed by $\sum d(n)n^{-3/2} < \infty$, permits both complete series to be opened and added separately.

Gaussian domination establishes the first integral's convergence; exponential domination establishes the second tail's convergence, and the oscillatory integral is over a finite interval. Fubini and the beta integral give $\mathcal{M}[K_0](s) = 2^{s-2}\Gamma(s/2)^2$ for $\operatorname{Re} s > 0$. For Y_0, we prove its Mellin identity first in the absolute strip $1/4 < \operatorname{Re} s < 1/2$ and identify the continued symbol by

$$\mathcal{M}[Y_0](2w) = -\frac{2}{\pi}\cos(\pi w)\mathcal{M}[K_0](2w).$$

Here the identity first holds where the Mellin integral of Y_0 converges. For subsequent line shifts, $\mathcal{M}[Y_0](s)$ denotes its explicit meromorphic Gamma-function continuation, not a divergent integral. To pair the actual Y_0 kernel with g, move the dual line to $\operatorname{Re} z = 13/16$, so $2(1-z)$ has real part $3/8$ inside that strip. Mellin pairing then gives $\mathcal{V}_-[g](n) =$

$-2\pi \int g(x) Y_0(4\pi\sqrt{nx})\,\mathrm{d}x$. Moving back is justified by holomorphy and proved decay; it is not an assumed identification of a divergent Mellin integral. The analogous K_0 pairing gives $\mathcal{V}_+[g](n) = 4\int g(x) K_0(4\pi\sqrt{nx})\,\mathrm{d}x$. Substitution into the two complete dual series proves (C.6). □

What is removed, and why. Apply Lemma C.1 to (C.3). The logarithmic main term is retained initially. The K_0 branch is bounded as a complete series using its exponential large-argument decay and the fact that its positive support is comparable to T. We prove a power saving $O_{\delta,A}(GT^{-A})$ for every fixed A; in the local-mean composition only the $A = 0$ consequence is needed. The main logarithmic integral is bounded by $O_\delta(G \log T)$ using its exact carrier phase and its smooth amplitude, not by replacing the divisor sum with an assumed asymptotic main term. Both bounds are absorbed into the existing error in (C.5).

C.4 The Bessel expansion and its stationary points

We prove the following two-term expansion for every $x > 0$:

$$Y_0(x) = \frac{\sqrt{\pi}}{\pi}\left((\sin x - \cos x)x^{-1/2} - \frac{\sin x + \cos x}{8}x^{-3/2}\right) + E_Y(x), \tag{C.7}$$
$$|E_Y(x)| \le \frac{2}{\pi}\frac{9}{128}\sqrt{\pi}\,x^{-5/2}.$$

It follows from the proved complex Laplace-ray representation of the literal Schläfli integral, a second-order amplitude remainder, and evaluation of the first two Laplace moments. Thus no large-argument Bessel expansion is left as a premise. Expressing the sine/cosine terms as exponentials retains two leading coefficients

$$c_+ = -(1+\mathrm{i})/2, \qquad c_- = -(1-\mathrm{i})/2$$

and two correction coefficients $(1-\mathrm{i})/2, (1+\mathrm{i})/2$, together with the $1/8$ factor. The complete correction and remainder series are proved absolutely integrable and are bounded by $O_\delta(G \log T)$ in the physical width range. Their terms are not silently discarded.

For a real signed frequency b, the leading continuous carrier is

$$\Phi(T, b; x) = T \log x - 2\pi x + 4\pi b\sqrt{x}.$$

Write $A = T/(2\pi)$ and

$$r_b = \frac{b + \sqrt{b^2 + 4A}}{2}, \qquad x_b = r_b^2.$$

For $T > 0$, $r_b > 0$ and $r_b^2 - b r_b = A$. Differentiation gives

$$\Phi_x = T/x - 2\pi + 2\pi b/\sqrt{x} = \frac{2\pi}{x}(A - x + b\sqrt{x}).$$

The positive solution of $x - b\sqrt{x} = A$ is uniquely x_b, so it is the unique positive saddle. Its second derivative is $-\pi(2r_b - b)/r_b^3 < 0$. The exact paired identities are

$$r_{-b} = r_b - b, \quad r_b r_{-b} = A, \quad x_b x_{-b} = A^2, \quad \log(x_b/A) = 2\,\mathrm{arsinh}\left(\frac{b}{2\sqrt{A}}\right).$$

All are obtained from the positive quadratic root and positive logarithm; there is no choice of a second branch.

The exact amplitude for power α is

$$a_\alpha(T, G, L; x) = x^{-\alpha}\Omega(\log x - \ell_T)\vartheta_{T,G,L}(x)\chi_T Q(T, G; \log x - \log A),$$

on positive support. Define its carrier integral by

$$\mathcal{A}_\alpha(T,G,L;b)=\int_0^\infty x^{-1/2}a_\alpha(T,G,L;x)e^{\mathrm{i}\Phi(T,b;x)}\,\mathrm{d}x.$$

The extra $x^{-1/2}$ belongs to the original divisor test; it is precisely canceled by the Jacobian under $x=y^2$. Put $h_b=1+A/r_b^2>1$. The stationary term is exactly

$$\mathcal{S}_\alpha(T,G,L;b)=2a_\alpha(T,G,L;x_b)e^{\mathrm{i}\Phi(T,b;x_b)}\frac{e^{-\mathrm{i}\pi/4}}{\sqrt{2h_b}}.$$

The 2 is the root-coordinate Jacobian combined with the root amplitude convention. The Fresnel value is proved from a damped Gaussian and an Abel limit; its tails are estimated before removing the damping.

Uniform stationary error above the fourth-root width.

Lemma C.2. *For each fixed real α there is $C_\alpha>0$ such that, if*

$$T\ge 1,\quad T^{1/4}\le G\le\sqrt{T},\quad L\ge 1,\quad 8L\le G,\quad |b|\le\sqrt{T}/100,$$

then

$$|\mathcal{A}_\alpha(T,G,L;b)-\mathcal{S}_\alpha(T,G,L;b)|\le C_\alpha G\sqrt{G}\,T^{-\alpha-1/4}.$$

The same estimate applies to both $b=\sqrt{n}$ and $b=-\sqrt{n}$ whenever $10000n\le T$.

Proof. Set $x=y^2$. The root phase is $2\pi(2A\log y-y^2+2by)$, with saddle r_b and quadratic part $\Phi(T,b;x_b)-2\pi h_b(y-r_b)^2$. We prove compact C^2 bounds for the root amplitude: its norm is at most a constant times $GT^{-\alpha}$, and first and second derivatives have scales $G/\sqrt{T}$ and $(G/\sqrt{T})^2$. These follow by differentiating the fixed Mellin profile, the smooth band cutoff, and the actual quadratic Gaussian; bounds remain uniform on the support, rather than depending on n or the chosen saddle.

For a symmetric radius h around r_b, the exact logarithmic Taylor expansion has cubic correction $2T(y-r_b)^3/(3r_b^3)$ and a bounded fourth-order remainder. Expand the amplitude to its linear term and the exponential phase to its cubic term. The linear amplitude and the cubic phase correction are odd around r_b, while the quadratic exponential is even; their two integrals vanish exactly. The remaining local error is bounded, with amplitude norm bound M and derivative scale R, by

$$M\left(2R^2h^3+4RTh^5/r_b^3+8Th^5/r_b^4+8T^2h^7/r_b^6\right).$$

This is obtained by Bounding these four residual terms before integration and multiplying by the interval length $2h$. The nonstationary portions are handled by the proved first-derivative/integration-by-parts estimates, and the finite quadratic window differs from its Fresnel value by an explicit reciprocal-radius tail.

For $|b|\le\sqrt{T}/100$, we prove $r_b\asymp\sqrt{T}$ with explicit window margins. Choose $X=T^{1/4}/\sqrt{G}$ and actual local radius $h=X/12$. The width assumptions give $X\ge 1$, $GX^2=\sqrt{T}$, $X^4=T/G^2\le\sqrt{T}$. After inserting $R=G/\sqrt{T}$, the combined reciprocal-radius and symmetric error is bounded by

$$\frac{4}{h\pi}+\frac{4G^2h^3}{T}+\frac{216Gh^5}{T}+\frac{1296h^5}{T}+\frac{11664h^7}{T}\le 20/X.$$

Each of the last four terms is bounded by $1/X$ from the displayed balances; the first is at most $16/X$ because $\pi>3$. Multiplying by $C_\alpha GT^{-\alpha}$ gives $C_\alpha GT^{-\alpha}/X=C_\alpha G\sqrt{G}T^{-\alpha-1/4}$. The small-frequency bound for both signs of $\sqrt{n}$ is exactly the consequence of $10000n\le T$. □

Define the complete stationary leading sum

$$\begin{aligned}\mathcal{S}(T,G,L) = \sum_{n\geq 1} d(n)(-2\pi)\frac{\sqrt{\pi}}{\pi}(4\pi)^{-1/2}n^{-1/4} \\ \times \Big(c_+\mathcal{S}_{1/4}(T,G,L;\sqrt{n}) + c_-\mathcal{S}_{1/4}(T,G,L;-\sqrt{n})\Big).\end{aligned} \tag{C.8}$$

For $8L \leq G$, both saddle amplitudes vanish when $n \geq N_*(T,G,L) = \lceil 36T(L/G)^2\rceil$; hence this sum is a finite sum with exactly the displayed terms. We prove the cutoff condition from the saddle logarithms and the support of ϑ, before taking the sum. The original leading carrier is also truncated at this cutoff with a proved tail bound. Lemma C.2 at $\alpha = 1/4$ and the divisor prefix bound $\sum_{n<N} d(n)n^{-1/4} \ll_\theta N^{3/4+\theta}$ give the finite sum error

$$O_\theta(G\sqrt{G}\,T^{-1/2}N_*^{3/4+\theta}).$$

When $G \geq T^{1/4+\kappa}$ and $L = \log T$, a smaller positive loss depending on δ, κ makes this $O(G)$; the strict κ margin absorbs the logarithms. We also prove that the complete two-term Neumann source differs from this complete stationary sum by $O_{\delta,\kappa}(G)$. Combining with (C.5) yields

$$I(T,G) \leq 2e\,\mathrm{Re}\,\mathcal{S}(T,G,\log T) + C_{\delta,\kappa}G\log T \tag{C.9}$$

for sufficiently large T and $T^\delta \leq G \leq T^{1/2-\delta}$, $G \geq T^{1/4+\kappa}$.

C.5 Signed weights and partial summation

The phase evaluated at the two saddles is expressed with

$$f(T,n) = 2T\,\mathrm{arsinh}\sqrt{\frac{\pi n}{2T}} + \sqrt{2\pi nT + \pi^2n^2} - \frac{\pi}{4}, \qquad c(T) = T\log(T/(2\pi)) - T.$$

It satisfies

$$\Phi(T,\sqrt{n};x_{\sqrt{n}}) = c(T) + \pi n + f(T,n) + \pi/4,$$

and the negative saddle gives $c(T) + \pi n - f(T,n) - \pi/4$. After multiplying by the Fresnel factor $e^{-\mathrm{i}\pi/4}$, the positive carrier is $e^{\mathrm{i}f}$ while the negative carrier is $-\mathrm{i}e^{-\mathrm{i}f}$. Neither is replaced by an unsigned absolute main term.

To display the full signed weights define

$$\begin{aligned}C_*(T,b) = \frac{1}{\sqrt{2}}(b^2 + 2T/\pi)^{-1/4}, \qquad c_n(T) = n^{-1/4}C_*(T,\sqrt{n}), \\ U(T,G;n) = Q\Big(T,G;2\,\mathrm{arsinh}\sqrt{\pi n/(2T)}\Big), \\ R_\pm(T,G,L;n) = \Omega(\log x_{\pm\sqrt{n}} - \ell_T)\vartheta_{T,G,L}(x_{\pm\sqrt{n}}), \\ w_\pm(T,G,L;n) = c_n(T)U(T,G;n)R_\pm(T,G,L;n), \\ E_\pm(T,n) = d(n)(-1)^n e^{\pm\mathrm{i}f(T,n)}.\end{aligned}$$

Then the exact identity for (C.8) is

$$\mathcal{S}(T,G,L) = (1+\mathrm{i})\chi_T e^{\mathrm{i}c(T)}\left(\sum_{n<N_*} w_+(T,G,L;n)E_+(T,n) - \sum_{n<N_*} w_-(T,G,L;n)E_-(T,n)\right). \tag{C.10}$$

It follows by algebraic substitution of the evaluated saddle formulas and the fourth-root normalization $2\sqrt{\pi}(4\pi)^{-1/2}n^{-1/4} = n^{-1/4}$. The two residual weights are different functions

at different saddles; $E_- = \overline{E_+}$ does not equate the weights. The common phase has modulus at most two.

For a finite sequence w_i, assume the uniform bound $|w_i| \le M$ for $0 \le i \le N$ and a separate bound $\sum_{i<N} |w_{i+1} - w_i| \le M$, with $M \ge 0$. For later notation, set

$$\mathcal{V}_{m,N}(w) = \max\left\{ \max_{0\le i\le N} |w(m+i)|, \sum_{i=0}^{N-1} |w(m+i+1) - w(m+i)| \right\}.$$

Thus the two bounds are equivalent to $\mathcal{V}_{m,N}(w) \le M$. This is a combined bound, not total variation alone. The finite summation-by-parts identity bounds $|\sum_{i<N} w_i E_i|$ by $2M$ times the maximum of the *actual* prefix sums $|\sum_{i<j} E_i|$, $0 \le j \le N$. For the two weights in (C.10), we prove the variation bounds

$$\mathcal{V}_{m,N}(w_\pm) \le CGT^{-1/4} m^{-1/4} e^{-G^2 m/(12T)}$$

when $m \ge 1$ and $10000(m+N) \le T$. The real fourth-root coefficient is positive antitone, so its variation telescopes. The Gaussian increment bound is derived by differentiating Q, and the two saddle residuals have uniform variation by differentiating the actual compact profiles. Multiplying the three variation bounds yields the displayed estimate.

Let

$$P_{m,N}(T) = \max_{0\le j\le N} \left| \sum_{i=0}^{j-1} d(m+i)(-1)^{m+i} e^{\mathrm{i}f(T,m+i)} \right|,$$

and define the undamped dyadic prefix bound

$$\mathscr{B}(T,N) = \sum_{0\le j<J(N)} (2^j)^{-1/4} P_{2^j,2^j}(T).$$

Here $J(N) = 0$ for $N \le 1$, and $J(N) = \lceil \log N/\log 2 \rceil$ for $N \ge 2$. This avoids taking a logarithm at $N = 0$. The empty sum and the zero-indexed stationary term are zero. Truncated dyadic pieces partition every finite prefix exactly. Summation by parts on each, then enlargement of its prefix maximum to the full block, gives

$$|\mathcal{S}(T,G,\log T)| \le C_\delta GT^{-1/4} \mathscr{B}(T, N_*(T,G,\log T)). \tag{C.11}$$

The damped factors are first kept; only their inequality $e^{-G^2m/(12T)} \le 1$ is used when passing to this undamped bound. Equations (C.9) and (C.11) give the required bound of a local second moment by maximal prefixes.

D Maximal prefixes and sums of local means

D.1 A Gram inequality for maximal prefixes

For a positive dyadic block $[m, m+N)$, the coefficients are $\alpha_i = d(m+i)(-1)^{m+i}$ and its unit phase vector at height t is $v_t(i) = e^{\mathrm{i}f(t,m+i)}$. At each height select a maximizing prefix length $j_t \in \{0,\dots,N\}$ for $P_{m,N}(t)$. Define the masked vector $v_t^{j_t}(i) = \mathbf{1}_{i<j_t} v_t(i)$. The exact Gram entry of two such vectors is

$$\sum_{i<N} \overline{v_t^{j_t}(i)} v_u^{j_u}(i) = \sum_{i<\min(j_t,j_u)} e^{\mathrm{i}(f(u,m+i)-f(t,m+i))}.$$

Thus each pair keeps the minimum of its two own maximizing prefixes. There is no single maximizing prefix imposed on all heights. If

$$G_{m,N}(t,u) = \max_{0\le j\le N} \left| \sum_{i<j} e^{\mathrm{i}(f(u,m+i)-f(t,m+i))} \right|, \quad E_{m,N} = \sum_{i<N} d(m+i)^2,$$

finite Hilbert-space Cauchy–Schwarz applied to the sum of phase-aligned masked vectors yields

$$\left(\sum_{t\in W} P_{m,N}(t)\right)^2 \le E_{m,N} \sum_{t,u\in W} G_{m,N}(t,u). \tag{D.1}$$

Indeed choose a unit scalar at each height to make its coefficient-vector inner product real and nonnegative. Cauchy–Schwarz bounds the square of their sum by the coefficient norm squared times the summed-vector norm squared; expanding that norm and taking absolute values gives (D.1). Zero prefixes or zero values cause no difficulty. We prove this finite lemma before using any analytic cancellation estimate.

Finite Cauchy–Schwarz over dyadic blocks gives, with $J = J(N)$,

$$\left(\sum_{t\in W} \mathscr{B}(t,N)\right)^2 \le J \sum_{j<J} (2^j)^{-1/2} E_{2^j,2^j} \sum_{t,u\in W} G_{2^j,2^j}(t,u). \tag{D.2}$$

The right side is the divisor-weighted sum of dyadic Gram entries. The divisor subpower estimate bounds $(2^j)^{-1/2}E_{2^j,2^j} \ll_\eta (2^j)^{1/2+\eta}$ for every $\eta > 0$.

D.2 First- and second-derivative bounds for each prefix

Treat the summation index as a positive real variable x. The exact phase slope and curvature are

$$f_x(T,x) = \sqrt{2\pi T/x + \pi^2}, \qquad f_{xx}(T,x) = -\frac{\pi T}{x^2\sqrt{2\pi T/x + \pi^2}}.$$

For $0 < x \le u < t \le 2u$, the phase difference $f(u,x) - f(t,x)$ has positive second derivative. Differentiating curvature with respect to height and integrating from u to t gives explicit upper and lower scales. On the entire block $m \le x \le m+N+2$ they are

$$\lambda = \frac{\pi(t-u)}{2(m+N+2)^2\sqrt{(2\pi+\pi^2)2u/m}},$$
$$\lambda' = \frac{\pi(t-u)}{m^2\sqrt{2\pi u/(m+N+2)}}.$$

Both are positive for $t > u$, $m \ge 1$. A second-order mean-value theorem puts every unit-spaced second difference between λ and λ'. The finite second-derivative estimate proved in Appendix F therefore bounds each prefix by

$$\left(\frac{N\lambda'}{2\pi} + 2\right)\left(\frac{2\pi}{\sqrt{\lambda}} + 2\left(\frac{\sqrt{\lambda}}{\lambda} + 1\right)\right).$$

A shorter prefix uses the same block curvature parameters; its length factor is merely enlarged to N. The zero prefix is handled separately.

For near height gaps, the first difference of $f(t,x) - f(u,x)$ is monotone and positive. The exact first-derivative scales place it in $[\delta_*, \pi]$ when $|t-u| \le \sqrt{Hm}$ and $t,u \in [H,2H]$, where $\delta_* > 0$ is the proved lower difference bound. The Kusmin–Landau one-period estimate gives $2\pi/\delta_*$. We take the minimum of this bound, the second-derivative estimate bound, and the triangle bound N. For the dyadic case $N = m$, its orientation-free majorant satisfies

$$G_{m,m}(t,t) = m, \tag{D.3}$$
$$G_{m,m}(t,u) \le 60\frac{\sqrt{Hm}}{|t-u|} \qquad (0 < |t-u| \le \sqrt{Hm}),$$

$$G_{m,m}(t,u) \le 2000\sqrt{\frac{m|t-u|}{\sqrt{Hm}}} \qquad (|t-u| \ge \sqrt{Hm}).$$

These statements require the block-plus-two endpoint at most the smaller height and the larger height at most twice the smaller. The physical cutoff guarantees both conditions simultaneously. In the far range the argument uses the triangle branch when curvature is large, so there is no extra linear gap term left in (D.3). More explicitly, put $d = |t-u|$, $R = \sqrt{Hm}$ and $Q = mR$. The actual curvature bounds give $\lambda \ge d/(320Q)$, $\lambda' \le 4d/Q$. If $R \le d \le Q$, the first second-derivative estimate factor is at most $6d/R$, while the second is at most $202\sqrt{Q/d}$. Their product is at most $1212\sqrt{md/R} \le 2000\sqrt{md/R}$. If $d > Q$, the triangle bound $m \le \sqrt{md/R}$ is stronger. These two cases prove the far estimate without an upper restriction on the gap.

A finite set is G-separated if distinct elements have distance at least G; here $G > 0$. The following estimate sums maximal prefixes over such ordinate sets.

Physical prefix sum bound.

Lemma D.1. *For $\delta, \eta > 0$ there are $C > 0$ and $H_0 \ge 40000$ such that, for $H \ge H_0$, $H^\delta \le G$, $L \ge 0$, a G-separated finite $W \subset [H, 2H] \cap [A, A+L]$, and $N = N_*(2H, G, \log(2H))$, $R = |W|$,*

$$\left(\sum_{t \in W} \mathscr{B}(t, N)\right)^2 \le C\mathcal{P}_\eta(H, G, L, N, R),$$

where $J_N = J(N)$ and

$$\mathcal{P}_\eta(H,G,L,N,R) = J_N^2 \Bigg[RN^{3/2+\eta} + \frac{120R\sqrt{H}}{G} N^{1+\eta} \mathsf{H}_{\lceil \sqrt{HN}/G \rceil} + 2000R^2 H^{-1/4} \sqrt{L}\, N^{3/4+\eta} \Bigg],$$

$$\mathsf{H}_k = \sum_{j=1}^{k} 1/j.$$

Proof. Eventually $1200 \log(2H) \le H^{\delta/2} \le G$ after the harmless doubled height adjustment. The explicit ceiling cutoff then satisfies $2N + 2 \le H$ and $N \le H$. Hence every dyadic block in (D.2) satisfies the prefix geometry needed for (D.3). The diagonal contributes Rm. For fixed t, order the G-separated points on either side of t. The jth has distance at least jG, so the near reciprocal sum is at most $120\sqrt{Hm}\,\mathsf{H}_{\lceil \sqrt{Hm}/G \rceil}/G$. Since every pair gap is at most L, the far double sum is at most $2000R^2\sqrt{mL/\sqrt{Hm}}$. Multiply these three contributions by the coefficient factor $Cm^{1/2+\eta}$. They become respectively $CRm^{3/2+\eta}$, $C(120R\sqrt{H}/G)m^{1+\eta}\mathsf{H}_{\lceil \sqrt{Hm}/G \rceil}$, and $C2000R^2H^{-1/4}\sqrt{L}m^{3/4+\eta}$. Each is nondecreasing with m on positive indices; use $m \le N$. The outer dyadic Cauchy–Schwarz factor and the number of blocks give $\lceil \log_2 N \rceil^2$. This proves the stated budget, including the empty-set case, for which all terms are zero. □

Sums of local second-moment excesses.

Proposition D.2. *For every $\delta, \kappa, \nu > 0$ there exist positive C, D and $H_0 \ge 40000$ such that the following holds. Let $H \ge H_0$, $G > 0$, $L \ge 0$ and a finite G-separated set W satisfy, for every $t \in W$,*

$$H \le t \le 2H, \quad t^\delta \le G \le t^{1/2-\delta}, \quad t^{1/4+\kappa} \le G, \quad A \le t \le A + L.$$

Set $E_G(t) = \max(0, I(t,G) - CG\log t)$. *Then*

$$\left(\sum_{t\in W} E_G(t)\right)^2 \le DH^\nu \left(\frac{|W|H}{G} + |W|^2\sqrt{GL}\right). \tag{D.4}$$

Proof. Equations (C.9) and (C.11) give $E_G(t) \le CGt^{-1/4}\mathscr{B}(t, N_*(t,G,\log t))$. The cutoff is monotone in $t \ge 1$ and the undamped prefix bound monotone in its cutoff. Thus $t \in [H, 2H]$ yields

$$E_G(t) \le CGH^{-1/4}\mathscr{B}(t,N), \qquad N = N_*(2H, G, \log(2H)).$$

Apply Lemma D.1 with loss $\nu/2$ and square the factor $CGH^{-1/4}$.

To evaluate the physical budget, the ceiling bound gives $N \le 74H(\log(2H))^2/G^2$, since $G \le \sqrt{2H}$. The diagonal index power then satisfies

$$G^2H^{-1/2}N^{3/2} \le 5476(H/G)(\log(2H))^3.$$

The near power becomes $(\sqrt{H}/G)(G^2H^{-1/2}N) = 74(H/G)(\log(2H))^2$. The harmonic factor is at most $2\log(2H)$ and the dyadic-count factor at most $((1/\log 2 + 1)\log(2H))^2$. The far power satisfies

$$H^{-1/4}\sqrt{L}\,G^2H^{-1/2}N^{3/4} \le 74\sqrt{GL}(\log(2H))^3.$$

The factor $N^{\nu/2} \le H^{\nu/2}$ and the remaining fifth power of the logarithm is eventually at most $H^{\nu/2}$. Substitution into all three terms proves (D.4). The constants are chosen before H, G, A, L, W; the analytic input is supplied by the stationary estimate. A nonempty sum gives $G \le \sqrt{2H}$ from one of its points; an empty sum has zero sum and is dealt with directly. □

E Large critical-line values and the twelfth moment

E.1 Absorption in a short physical interval

Finite local-to-global absorption.

Lemma E.1. *Let* $A, H, G, Y > 0$, *and a finite* $W \subset [H, 2H]$ *have nonnegative values* $f(t) \ge Y$. *Suppose every* $U \subset W$ *localized in an interval of length*

$$L_* = (Y^2/(4A))^2/G$$

satisfies $\left(\sum_{t\in U} f(t)\right)^2 \le A(|U|H/G + |U|^2\sqrt{GL_*})$. *Then*

$$|W| \le \frac{2AH}{GY^2} + \frac{32A^3H^2}{Y^6}.$$

Proof. $\sqrt{GL_*} = Y^2/(4A)$, so $2A\sqrt{GL_*} \le Y^2$. For $R = |U| > 0$, use $\sum f \ge RY$ in the local sum and absorb its quadratic term to get $R \le 2AH/(GY^2)$. The empty fiber also satisfies this. Partition $[H, 2H]$ by the natural floor bins of $(t-H)/L_*$; at most $\lfloor H/L_* \rfloor + 1$ fibers are occupied. This convention also captures a point on the closed right boundary. Sum the fiber cardinality bounds and use $\lfloor H/L_* \rfloor \le H/L_*$. Expanding gives exactly $2AH/(GY^2) + 32A^3H^2/Y^6$. □

Applying this lemma to Proposition D.2, with its loss $\nu/3$, yields the actual local-mean-excess count

$$|W| \le DH^\nu \left(\frac{H}{GY^2} + \frac{H^2}{Y^6}\right) \quad (E_G(t) \ge Y). \tag{E.1}$$

The cube of $H^{\nu/3}$ is H^ν; the first term is enlarged to that same loss. Thus no unseen exponent is spent in the global height cover.

E.2 Clustering separated large values

Let $W \subset [H, 2H]$ be one-separated and satisfy $Z(t) \geq V > 0$. For a width G, partition by $n = \lfloor (t-H)/(G/2) \rfloor$. The center assigned to bin n is its *left* endpoint $c_n = H + nG/2$, not its midpoint. Its cluster lies in $[c_n, c_n + G/2]$, hence also in $[c_n - G/2, c_n + G/2]$. The nonempty cluster centers remain in $[H, 2H]$. Centers with equal parity are G-separated; split into even and odd bins before using (E.1).

Set $\ell = \log(3H)$, and let $P > 0$ be the point-mean constant in Lemma B.2. If $2\ell^2 \leq G \leq H$, that lemma gives, for cluster occupancy r_n,

$$V^2 r_n \leq P\ell(r_n + 4I(c_n, G)).$$

If $V^2 \geq 2P\ell$, then

$$I(c_n, G) \geq 2r_n A_V, \qquad A_V = V^2/(16P\ell).$$

If the local-mean error $CG\ell \leq A_V$, then for every integer $m \geq 1$ with $r_n \geq m$, the excess at c_n is at least mA_V. Apply (E.1) to the two parity classes of centers. The result is

$$\#\{n : r_n \geq m\} \leq 2DH^\nu \left(\frac{H}{G(mA_V)^2} + \frac{H^2}{(mA_V)^6} \right) \leq \frac{2DH^\nu}{m^2} \left(\frac{H}{GA_V^2} + \frac{H^2}{A_V^6} \right).$$

The exact integer layer-cake identity $\sum_n r_n = \sum_{m \geq 1} \#\{n : r_n \geq m\}$ and $\sum_{m \geq 1} m^{-2} \leq 2$ therefore give

$$|W| \leq D' H^\nu \left(\frac{H}{GA_V^2} + \frac{H^2}{A_V^6} \right). \tag{E.2}$$

This occupancy step restores *all* peaks, not merely one representative per cluster.

Choose $K = 16PC + 1$ and

$$G = \frac{V^2}{K\ell^2}.$$

The error absorption is automatic: $CG\ell \leq V^2/(16P\ell) = A_V$. When

$$K\ell^2 (2H)^{1/4+\kappa} \leq V^2 \leq K\ell^2 H^{1/2-\delta}, \qquad 0 < \delta \leq 1/4,$$

all physical width conditions hold simultaneously throughout $[H, 2H]$; the lower bound also makes $2\ell^2 \leq G$ eventually. Substitution into (E.2) gives

$$|W| \leq DH^\nu \left(\frac{H\ell^4}{V^6} + \frac{H^2\ell^6}{V^{12}} \right). \tag{E.3}$$

The factors of K and $16P$ are explicitly absorbed into D by direct substitution.

High-value finite-set bound.

Proposition E.2. *For every $\varepsilon > 0$, eventually in H, for all $V > 0$ with $V \geq H^{1/8+\varepsilon}$ and every one-separated finite $W \subset \{t \in [H, 2H] : Z(t) \geq V\}$,*

$$|W| V^{12} \leq H^{2+\varepsilon}.$$

Proof. Choose $\eta = \min(\varepsilon/16, 1/1000)$, use $\delta = 1/48$, $\kappa = \eta$, $\nu = \eta$ in (E.3). The lower source range follows eventually from $V^2 \geq H^{1/4+2\varepsilon} \geq H^{1/4+2\eta}$, because the extra positive η power absorbs $K\ell^2 2^{1/4+\eta}$. If W is empty there is nothing to prove. Otherwise take a point in W. Proposition 3.1 gives $V \leq H^{1/6+\eta}$. Consequently $V^2 \leq H^{1/3+2\eta} \leq K\ell^2 H^{1/2-1/48}$ eventually, so the upper source range also holds uniformly. Multiply (E.3) by V^{12}:

$$|W| V^{12} \leq DH^\eta (H\ell^4 V^6 + H^2\ell^6) \leq D\ell^4 H^{2+7\eta} + D\ell^6 H^{2+\eta}.$$

Each logarithm and constant is absorbed into the spare positive loss, giving $H^{2+8\eta} \le H^{2+\varepsilon}$. The choices and thresholds depend on ε and the two analytic estimates, but precede H, V, W. Substituting the proved point-mean and stationary sum estimates removes both explicit input hypotheses. □

From every finite separated set to full measure.

Lemma E.3. *Put $S(H,V) = \{t : H \le t \le 2H,\ Z(t) \ge V\}$. Under the eventual conditions of Proposition E.2,*

$$|S(H,V)|_{\mathrm{Leb}} \le 2H^{2+\varepsilon}/V^{12}.$$

Proof. A uniform cardinality bound $|W| \le B$ on every one-separated finite subset of a set S implies that the possible cardinalities have a largest value: they lie in the finite set $\{0, \dots, \lceil B \rceil\}$. Choose a one-separated subset of that maximal size. If a point of S were farther than one from every chosen point, it could be inserted to increase the size. Thus closed radius-one balls about the chosen points cover S. Their total length is at most $2B$. Take $B = H^{2+\varepsilon}/V^{12}$. Continuity of Z makes $S(H,V)$ closed and measurable. This proves a measure bound for the full superlevel set, not just for a sampled subset. □

E.3 Layer cake, the high/low split and the physical interval

Truncated high moment.

Lemma E.4. *Suppose $V > 0$, $V^{12} \le M$, $C \ge 0$, $Z(t)^{12} \le M$ on $S(H,V)$, and $|S(H,U)|_{\mathrm{Leb}} \le C/U^{12}$ for every $U \ge V$. Then*

$$\int_{S(H,V)} Z(t)^{12}\,\mathrm{d}t \le C(1 + \log(M/V^{12})).$$

Proof. For the restriction to $S(H,V)$ the distribution of $f(t) = Z(t)^{12}$ is bounded by $C/\max(V^{12}, s)$ at level $s > 0$. If $s \le V^{12}$ use the whole-set bound; if $s > V^{12}$ use the level $U = s^{1/12} \ge V$. Positivity and monotonicity of the twelfth power justify this change of level. The function is integrable on this bounded measurable set by continuity of Z. Apply layer cake up to M and integrate:

$$\int_0^M \frac{C}{\max(V^{12}, s)}\,\mathrm{d}s = C + C\int_{V^{12}}^M \frac{\mathrm{d}s}{s} = C(1 + \log(M/V^{12})).$$

□

We now reconstruct the proof of Theorem 1.3.

Proof. First prove the dyadic bound. Given $\varepsilon > 0$ put $q = \varepsilon/20$. Proposition A.3 gives $\int_H^{2H} Z^4 \le CH^{1+q}$ eventually. For the high-moment step use a smaller loss $\eta = \min(q/2, 1/100)$ in Lemma E.3, start at $V_0 = H^{1/8+\eta}$ and use the coarse upper truncation $M = H^3$ from Proposition 3.1. Then $1 \le V_0^{12} \le H^3$ for $H \ge 1$, and Lemma E.4 gives

$$\int_{S(H,V_0)} Z(t)^{12}\,\mathrm{d}t \le 2H^{2+\eta}(1 + \log(H^3/V_0^{12})) \le 2H^{2+\eta}(1 + 3\log H) \le H^{2+q}$$

eventually, since $\eta < q$. The set $S(H, H^{1/8+q})$ is a subset of this starting high set, so its integral obeys the same bound.

On the complement of $S(H, H^{1/8+q})$ within $[H, 2H]$, $Z^{12} \le H^{8(1/8+q)} Z^4$. Thus

$$\begin{aligned}\int_H^{2H} Z(t)^{12}\,\mathrm{d}t &\le H^{2+q} + H^{1+8q}\int_H^{2H} Z(t)^4\,\mathrm{d}t \\ &\le H^{2+q} + CH^{2+9q} \le (1+C)H^{2+\varepsilon}.\end{aligned}$$

This is the dyadic estimate, with a positive constant $1 + C$. All integrals over the high set and its complement are finite because they are restrictions of a continuous function on a compact interval.

To pass to $[0, 3T]$, let $B \ge 1$ exceed the dyadic threshold and set $r = 2 + \varepsilon \ge 1$. For $f = Z^{12}$ choose $A = |\int_0^B f| + (1+C) + 1 > 0$, so the dyadic constant $1 + C \le A$. Continuity proves that the compact initial integral is finite, and $\int_0^B f \le AB^r$. If $\int_0^{2^m B} f \le A(2^m B)^r$, split at $2^m B$ and apply the dyadic bound to the next interval. The sum is at most $2A(2^m B)^r \le A(2^{m+1}B)^r$, because $2 \le 2^r$. This induction is exactly the zero-origin adapter. For $X \ge B$, choose $X \le 2^m B \le 2X$. Nonnegativity permits enlargement to this dyadic endpoint, giving $\int_0^X f \le A2^r X^r$ with no further loss. Put $X = 3T$ and use $(3T)^r = 3^r T^r$; choose $K = A2^r 3^r > 0$. The identity between interval and closed-set integrals gives the literal $\int_{[0,3T]}$ endpoint. Finally continuity of $s(t)$ and analyticity of ζ away from 1 show Z continuous on $\mathbb{R}$, hence all its natural powers are integrable on every finite $[a, b]$. This completes the reconstruction of the moment estimate. □

F Finite exponential sums and analytic details

This appendix expands selected analytic adapters whose exact construction is important for reading the main proof. They are locally proved results, not extra assumptions. All branches and finite-limit conventions here agree with the production imports.

F.1 Weyl growth and a sharp approximate functional equation

For $s \in \mathbb{C}$ and a real cutoff x, let $S(s, x) = \sum_{1 \le n \le \lfloor x \rfloor_+} n^{-s}$ and define the literal remainder

$$R_{\mathrm{AFE}}(s; x, y) = \zeta(s) - S(s, x) - \chi(s)S(1 - s, y), \quad \chi(s) = 2^s \pi^{s-1}\Gamma(1 - s)\sin(\pi s/2).$$

The unit term is included in both sums. Defining this remainder alone supplies no estimate on it; the bounded-error estimate is the following.

For $t \ge 1$, $s = 1/2 + \mathrm{i}t$, assume

$$\begin{aligned} &x, y \ge 1, \quad y \le x, \quad 2\pi xy = t, \quad x = k + 1/2\ (k \in \mathbb{Z}), \\ &\lfloor y \rfloor_+ + 1/8 \le y \le \lfloor y \rfloor_+ + 1/2, \qquad e^{-\pi t}(t^2 + 1) \le 1. \end{aligned}$$

Then $|R_{\mathrm{AFE}}(s; x, y)| \le C_{\mathrm{AFE}}$, where the calculation gives the explicit positive constant

$$\begin{aligned} C_P &= \frac{\log 2 + 1}{2\pi} + \frac{1 + 2\pi}{4\pi^2}(\pi/2 + \log 2) + \frac{2(1 + 2\pi)}{4\pi^3}\,9 + \frac{1 + 2\pi}{4\pi^3}\,17, \\ C_{\mathrm{AFE}} &= C_P + 3 + 24/\pi + (1 - e^{-\pi})^{-1}. \end{aligned}$$

Here is the decomposition and error proof. Define the actual finite oscillatory integral

$$J(s; a, b; m) = \int_a^b u^{-s} e^{2\pi \mathrm{i} m u}\,\mathrm{d}u$$

and $J_\infty(s; a; m)$ as the limit as $b \to \infty$ of this integral for $0 < \mathrm{Re}\, s < 1$, $m > 0$. It is not an undamped whole-half-line Lebesgue integral. The upper-tail integrals are defined initially

for $a > 0$; at $a = 0$, $J_\infty(s; 0; m)$ means the limit of the improper integral from zero, which converges at its lower endpoint when $\operatorname{Re} s < 1$. Let $M = \lfloor y \rfloor_+$ and

$$B = \frac{x^{1-s}}{s-1}, \qquad F = \sum_{m=1}^{M} J_\infty(s; x; m), \qquad F_0 = \sum_{m=1}^{M} J_\infty(s; 0; m),$$
$$L = \sum_{m=1}^{M} J(s; 0, x; m), \qquad Q = \chi(s) S(1-s, y).$$

The Abel sawtooth identity for actual zeta, followed by its finite Poisson expansion and the proved limit of its positive and negative frequency tails, gives

$$|\zeta(s) - S(s, x) - B - F| \le C_P.$$

For clarity, the four contributions to C_P are the endpoint first-order coefficient, the paired harmonic endpoint coefficient, and the two higher coefficient tails. On the half-integral x, alternating harmonic tails have their explicit bounds. The first coefficient bounds are denoted by $C_2^+(y), C_3^+(y)$, and the upper-frequency coefficient bounds by $C_2^-(M, y), C_3^-(M, y)$. Explicitly, with $d_y = M + 1 - y$, they are

$$C_2^+(y) = \frac{\log(y+1) + \gamma_E}{y^2} - \frac{1 + 2y}{2y^2(y+1)},$$
$$C_3^+(y) = \frac{\log(y+1) + \gamma_E}{y^3} - \frac{1 + 3y + 3y^2}{2y^3(y+1)^2},$$
$$C_2^-(M, y) = \frac{d_y^{-2} + (d_y+1)^{-2} + (d_y+1)^{-1}}{y} - \frac{\log(M+1) - (M+1)^{-1} - \operatorname{Re}\psi_\Gamma(d_y)}{y^2},$$
$$C_3^-(M, y) = \frac{d_y^{-3} + (d_y+1)^{-3} + [2(d_y+1)^2]^{-1}}{y} - \frac{d_y^{-2} + (d_y+1)^{-1}}{y^2} + \frac{\log(M+1) - [2(M+1)]^{-1} - \operatorname{Re}\psi_\Gamma(d_y)}{y^3}.$$

Under the cutoff gaps, the coefficient estimates give

$$C_2^-(M, y) + C_2^+(y) \le 9/y, \qquad C_3^-(M, y) + C_3^+(y) \le 17/y.$$

The differentiated endpoint amplitudes are

$$H = (1/2 + t)x^{-3/2}, \qquad H_1 = (3/2)(1/2 + t)x^{-5/2}.$$

They arise from $g(u) = u^{-1/2}$ and $f(u) = (t/(2\pi)) \log u$ through $H = |g'(x)| + 2\pi|g(x)f'(x)|$ and $H_1 = |g''(x)| + 2\pi(|g(x)f''(x)| + |g'(x)f'(x)|)$. They satisfy $H/y \le 1 + 2\pi$, $H_1/y \le 2(1 + 2\pi)$ and $H(y/x)/y \le 1 + 2\pi$, because $y \le x$ and $x \ge 1$. Substitution into the four terms yields exactly C_P.

The elementary pole term obeys $|B| \le 1$. For the lower integrals, integration by parts in their logarithmic phase and retention of the half-integer endpoint produce the arithmetic gap sum

$$\sum_{m=1}^{M} \frac{m^2}{y^2(y-m)} \le 8(1 + \log y).$$

The gap $y - M \ge 1/8$ controls the closest term; for the other terms the harmonic sum comparison gives the logarithm. Since $x^{-1/2}(1 + \log y) \le 3$ for $1 \le y \le x$, and the two boundary factors are each at most one, this proves $|L| \le 2 + 24/\pi$.

To evaluate F_0, introduce positive damping $e^{-\epsilon u}$. The convergent Gamma–Laplace integral is

$$\int_0^\infty u^{-s} e^{2\pi i m u - \epsilon u}\,\mathrm{d}u = (\epsilon - 2\pi i m)^{s-1}\Gamma(1-s), \qquad \epsilon > 0.$$

On any finite interval, dominated convergence removes the damping. On an upper interval starting at $R > t/(\pi m)$, the phase derivative has magnitude at least πm, and the nonnegative amplitude $u^{-\operatorname{Re} s} e^{-\epsilon u}$ is decreasing. The proved primitive bound and integration by parts give the uniform tail bound $2R^{-\operatorname{Re} s}/(\pi m)$. Thus the finite-interval Abel limit and this tail bound identify

$$J_\infty(s;0;m) = (-2\pi i m)^{s-1}\Gamma(1-s).$$

The negative-imaginary-axis principal branch is retained. Its exact factorization is

$$\Gamma(1-s)(2\pi/\mathrm{i})^{s-1} = \chi(s)(1+e_\Gamma(s)), \qquad e_\Gamma(s) = \frac{e^{\pi i s}}{1-e^{\pi i s}}.$$

The denominator has norm at least $1-e^{-\pi|t|}$; hence for $t \ge 1$, $|e_\Gamma(s)| \le e^{-\pi t}/(1-e^{-\pi})$. Also $|S(1-s,y)| \le y^{1/2}\log y + 1 \le t^2 + 1$. Since $|\chi(s)| = 1$, this gives $|F_0 - Q| \le (1-e^{-\pi})^{-1}$. The exact relation $F = F_0 - L$ and

$$R_{\mathrm{AFE}} = (\zeta - S - B - F) + B + (F - Q)$$

prove the displayed constant bound. This makes all three analytic error sources explicit, rather than postulating a sharp functional equation.

Finally we construct the cutoffs at every $t \ge 200\pi$. Let $r = \sqrt{t/(2\pi)} \ge 10$, set

$$M = \lfloor r/2 \rfloor_+, \quad b = M + 1/2, \quad u = r^2/b, \quad A = \lfloor u + 1/2 \rfloor_+, \quad x = A + 1/2, \quad y = r^2/x.$$

The floor inequalities give $u < x \le u + 1$, and consequently

$$r \le x \le (12/5)r, \quad (2/5)r \le y \le r, \quad M + 1/8 \le y \le M + 1/2, \quad \lfloor y \rfloor_+ = M.$$

The first scale range and $2\pi > 6$ give $A^2 \le t$; the second gives $M^2 \le t$. The exponential envelope is eventually at most one. These are precisely the short-polynomial cutoffs used in Proposition 3.1.

F.2 A Laplace proof of the Bessel remainder

The exact ray proof of (C.7) is short enough to give explicitly. Define, with principal half-powers,

$$K_x(z) = e^{\mathrm{i}xz}(1-z)^{-1/2}(1+z)^{-1/2}, \qquad V_x(b) = \int_0^\infty K_x(b + \mathrm{i}t)\,\mathrm{d}t.$$

For $0 \le b < 1$ a rectangle in the upper half-plane, its upper edge bounded by C/H times e^{-xH}, yields the equality of the two vertical integrals and the finite real edge. Under $z = \sin u$, the positive cosines cancel the two principal half-powers, giving

$$\int_0^{\pi/2} e^{\mathrm{i}x\cos u}\,\mathrm{d}u = \mathrm{i}V_x(0) - \mathrm{i}V_x(1).$$

We take $b = \sin(\pi/2 - H^{-1}) \to 1$ and prove convergence of the vertical integral at $b = 1$, where the local endpoint majorant is $t^{-1/2}$. The $b = 0$ integral is the real tail $\int_0^\infty e^{-xt}/\sqrt{1+t^2}\,\mathrm{d}t$. Taking imaginary parts and comparing with the Schläfli definition gives $Y_0(x) = -(2/\pi)\operatorname{Re} V_x(1)$.

On the unit ray,

$$V_x(1) = e^{\mathrm{i}x}\frac{1+\mathrm{i}}{2}\int_0^\infty t^{-1/2}e^{-xt}(1+\mathrm{i}t/2)^{-1/2}\,\mathrm{d}t.$$

This factorization uses multiplication of a positive real base with a nonzero complex base, which does not cross the principal logarithm's cut. The constant is exactly $(-\mathrm{i})^{-1/2}2^{-1/2} = (1+\mathrm{i})/2$. Put $w = (1+\mathrm{i}u)^{1/2}$. Then $\operatorname{Re} w \ge 1$, $|w| \ge 1$, $|w+1| \ge 2$, and $w - 1 = \mathrm{i}u/(w+1)$. The algebraic identity

$$2w(w^{-1} - 1 + \mathrm{i}u/2) = (w-1)^2(w+2)$$

therefore implies the global bound

$$|(1+\mathrm{i}u)^{-1/2} - 1 + \mathrm{i}u/2| \le 3u^2/8.$$

At $u = t/2$, multiply by $t^{-1/2}e^{-xt}$ and integrate. The three exact moments are

$$\int_0^\infty t^{k-1/2}e^{-xt}\,\mathrm{d}t = \Gamma(k+1/2)x^{-k-1/2}, \quad k = 0, 1, 2.$$

Their values are $\sqrt{\pi}x^{-1/2}$, $(\sqrt{\pi}/2)x^{-3/2}$ and $(3\sqrt{\pi}/4)x^{-5/2}$. Thus the Laplace integral differs from $\sqrt{\pi}x^{-1/2} - \mathrm{i}\sqrt{\pi}x^{-3/2}/8$ by at most $(9/128)\sqrt{\pi}x^{-5/2}$. Multiplication by $e^{\mathrm{i}x}(1+\mathrm{i})/2$, taking its real part, and multiplying by $-2/\pi$ give (C.7) and its stated error. All these integrals are absolutely convergent for $x > 0$; no asymptotic series or unproved Bessel property enters the computation.

F.3 Finite Fresnel windows and the symmetric stationary cancellation

For $c > 0$, let $F_c(H) = \int_{-H}^{H} e^{-2\pi\mathrm{i}cz^2}\,\mathrm{d}z$. For positive damping, the whole-line Gaussian value is $G_\epsilon(c) = (\pi/(\epsilon + 2\pi c\mathrm{i}))^{1/2}$. The phase derivative outside a window of radius H is at least $4\pi cH$; using its explicit reciprocal derivative, integration by parts on each tail bounds the difference between $G_\epsilon(c)$ and its damped window uniformly in $\epsilon > 0$ by $2/(cH\pi)$. On the finite interval the damped integrand converges continuously to the undamped one. The principal half-power is continuous at $\pi/(2\pi c\mathrm{i}) = -\mathrm{i}/(2c)$, so

$$|F_c(H) - (-\mathrm{i}/(2c))^{1/2}| \le 2/(cH\pi).$$

The number $(1-\mathrm{i})/(2\sqrt{c})$ has positive real part and square $-\mathrm{i}/(2c)$, hence is that principal square root. Equivalently it is $e^{-\mathrm{i}\pi/4}/\sqrt{2c}$, proving the exact Fresnel normalization and its finite-window error.

For the stationary integral, write $y = r + z$ with $r = r_b$ and $\Theta(z) = \Phi(T, b; (r+z)^2)$. For $|z| \le r/2$, the actual logarithm expansion gives

$$\Theta(z) = \Theta(0) - 2\pi h_b z^2 + \frac{2T}{3r^3}z^3 + E_4(z), \qquad |E_4(z)| \le 4T|z|^4/r^4.$$

This follows from the logarithmic power-series remainder, multiplied by $2T$, and the exact quadratic-root equation canceling its linear term. If the root amplitude is $a(r+z)$, its C^2 bound gives $|a(r+z) - a(r) - za'(r)| \le MR^2z^2$. The cubic correction multiplied by $a(r)$ is odd, as is $za'(r)$; both integrate to zero against the even quadratic exponential on $[-h, h]$. The remaining terms are: the quadratic amplitude remainder; the product of linear amplitude with cubic phase; the fourth-order phase remainder times the bounded amplitude; and the

second-order exponential remainder of the cubic term. Bounding these four terms before integration gives respectively

$$2MR^2h^3, \quad 4MRTh^5/r^3, \quad 8MTh^5/r^4, \quad 8MT^2h^7/r^6.$$

This is the local symmetric estimate in Lemma C.2. The tails use actual compact support, vanishing endpoint amplitudes, and the nonstationary primitive estimates; the Fresnel-window bound then provides the reciprocal h term. Hence the fourth-root balance follows from an exact odd-term cancellation, not from a generic stationary-phase theorem with an unspecified remainder.

F.4 Finite cancellation lemmas used twice in the development

The Weyl estimate and the maximal-prefix argument both use the following finite cancellation mechanism. It is the discrete first-derivative test, followed by a second-derivative estimate. Let $\theta_n = f(n+1) - f(n)$ be monotone with $\delta \le \theta_n \le 2\pi - \delta$, $\delta > 0$, $0 \le n \le N$. The phase resolvent

$$q(\theta) = \frac{1}{1 - e^{\mathrm{i}\theta}} = \frac{1}{2} + \frac{\mathrm{i}}{2}\cot(\theta/2)$$

has constant real part. The identity $e^{\mathrm{i}f(n)} = q(\theta_n)(e^{\mathrm{i}f(n)} - e^{\mathrm{i}f(n+1)})$ telescopes by finite summation by parts. Its resulting norm bound is the two endpoint resolvent norms plus the variation of $q(\theta_n)$. Since cot is decreasing on $(0, \pi)$, that variation is $(\cot(\theta_0/2) - \cot(\theta_N/2))/2$. Each endpoint satisfies $|q(\theta)| \le \pi/(2\delta)$; its cotangent absolute value divided by two is at most $|q(\theta)|$. Thus the phase sum is at most $2\pi/\delta$. Subtracting $2\pi kn$ leaves every phase unchanged and reduces any fixed period to this case. Replacing $f(n)$ by $-f(n) + 2\pi n$ gives the decreasing increment version by complex conjugation.

If instead the successive second differences satisfy $0 < \lambda \le \theta_{n+1} - \theta_n \le \Lambda$ for $n < N$, partition by $k_n = \lfloor \theta_n/(2\pi) \rfloor$. The index is monotone. The number of periods from k_0 to k_N is at most $N\Lambda/(2\pi) + 2$, since $\theta_N - \theta_0 \le N\Lambda$. In each period the good increments lie in $[2\pi k + \delta, 2\pi(k+1) - \delta]$ and form a consecutive index set; their phase sum is at most $2\pi/\delta$ by the resolvent estimate. The two remaining endpoint bands have at most $\delta/\lambda + 1$ indices each: if i and j are in a band of width δ, then $(j - i)\lambda \le \theta_j - \theta_i \le \delta$. Absolute estimation on those bands gives

$$\left| \sum_{n=0}^{N} e^{\mathrm{i}f(n)} \right| \le \left(\frac{N\Lambda}{2\pi} + 2 \right) \left(\frac{2\pi}{\delta} + 2(\delta/\lambda + 1) \right).$$

This estimate remains valid if the good band is empty or the two excluded bands overlap, since the partition inequalities are upper bounds. Taking $\delta = \sqrt{\lambda}$ proves the second-derivative estimate used in Section D with its exact constants.

For completeness, the finite Weyl A-process is also an exact sum argument. Extend a coefficient sequence by zero off $[0, N)$ and average its H shifts on $[-H, N)$. Each original term appears H times. Cauchy–Schwarz on this padded interval gives

$$H^2 \left| \sum_{n<N} a_n \right|^2 \le (N + H) \left(HN + H^2 C \right)$$

when $|a_n| \le 1$ and every off-diagonal shifted correlation has norm at most C. Expansion of the square supplies HN from the H diagonal shifts; the other correlations have real part at most their norm, so their total is at most H^2C. For the logarithmic phase $f(x) = -t \log x$ on a block starting at A, the shift difference has exact curvature $t/u^2 - t/v^2 = tr(u+v)/(u^2v^2)$, where $v - u = r > 0$ and $A \le u \le v \le 2A$. Hence its unit second differences lie between

$tr/(8A^3)$ and $4tr/A^3$. The preceding second-derivative estimate and the elementary product estimate, when $tr/(8A^3) \le 1$, give the correlation bound

$$100\left(N\sqrt{tr/(8A^3)} + 1/\sqrt{tr/(8A^3)}\right).$$

Uniformize over $1 \le r \le H$ by replacing the first square root with its value at H and the second denominator with its value at one. For $t = Y^3$, $1 \le Y \le A$, $A^2 \le Y^3$, choose $H = \lfloor A/Y \rfloor_+$. Then $1 \le H \le A$, $HY \le A \le 2HY$. The uniform correlation is at most $400Y$. With outer block length A, the A-process gives a square bound at most $900AY$, hence phase-prefix norm at most $30\sqrt{AY}$. For shorter prefixes, if their length is below H, use the triangle bound; otherwise apply the same shift argument and enlarge its length factors to A. Thus the prefix bound is genuinely uniform over all prefix lengths.

Finally for any nonnegative antitone weight u_n, summation by parts gives

$$\left|\sum_{n<N} u_n v_n\right| \le B\left(u_{N-1} + \sum_{n<N-1}(u_n - u_{n+1})\right) = Bu_0$$

if every phase prefix has norm at most B. Take $u_n = (A+n)^{-1/2}$ and $B = 30\sqrt{AY}$ to get $30\sqrt{Y} = 30t^{1/6}$, the physical block estimate in Proposition 3.1.

F.5 Summable nonstationary terms

Here are the concrete series majorants behind the removals in Section C. On the support $x \in [T/16, T]$, the actual K_0 integral has exponential large-argument domination. For every fixed integer k, its pointwise bound is at most $C_k/(T^k n^k)$ at $4\pi\sqrt{nx}$. We first use $\cosh u \ge 1 + u^2/2$ to prove $|K_0(z)| \le 2/\sqrt{z}$ for $z > 0$. Splitting the exponential as two equal factors gives $K_0(z) \le e^{-z/2}K_0(z/2)$, hence $|K_0(z)| \le 2e^{-z/2}$ for $z \ge 2$. Since $e^{-z/2} \le (2k)!/(z/2)^{2k}$, and $(4\pi\sqrt{nx})^2 \ge Tn$ on the support, the power constant can be chosen $C_k = 2 \cdot 2^{2k}(2k)!$. The compact source test has integral norm at most CGT. Therefore its nth full plus term is at most

$$\frac{C_k GT}{T^k} d(n) n^{-2} \qquad (k \ge 2).$$

The zero term vanishes. Since $\sum d(n)n^{-2} < \infty$, summing proves $|\text{plus branch}| \le C_k GT/T^k$. Choose $k > A + 2$ and then $T \ge C_k$ to obtain the power saving GT^{-A}.

The correction carriers in (C.7) have $\alpha = 3/4$. The uniform root-kernel primitive bound is at most four, obtained by splitting at $r_b - 1, r_b + 1$ and applying the one-sided nonstationary derivative bound outside the middle interval of length two. Integration by parts with the actual C^1 amplitude gives $|\mathcal{A}_\alpha(T, G, L; b)| \le C_\alpha GT^{-\alpha}$ for every real b. For $b \ge 8\sqrt{T}$, the support is away from both signed saddles and the stronger bound is $C_\alpha GT^{-\alpha}/b$ for each sign. Combining these two bounds at $b = \sqrt{n}$, using $T^{-3/4}\sqrt{T} \le 1$, gives a correction pair bounded by $CG/\sqrt{n}$. Its explicit Bessel factor $(4\pi)^{-3/2}n^{-3/4}/8$ and divisor factor then give $CGd(n)n^{-5/4}$ per term. The convergent divisor series at $5/4$ bounds the complete correction by $O(G)$.

The genuine remainder in (C.7) is bounded on the same support by a constant times $T^{-5/4}n^{-5/4}$. Multiply by the compact-test integral norm CGT and sum against $d(n)$; the resulting bound is $CGT^{-1/4}\sum d(n)n^{-5/4} \le C'G$. These are full-series bounds, so neither correction nor remainder can hide an unresolved infinite tail.

The logarithmic main term is written as a smooth weighted reflection integral with kernel $x^{-1}e^{\mathrm{i}(T\log x - 2\pi x)}$ on $[T/16, T]$. Its C^1 weight bound is $CG\sqrt{T}\log T$. The all-frequency estimate in Lemma 4.4, followed by integration by parts with this weight, cancels the $\sqrt{T}$

scale and bounds the main term by $C'G \log T$. This is why that term is absorbed into the local-mean error rather than set to zero. The fixed support, coefficient norms, derivative integrals and primitive estimates all enter before physical scales are specialized to $L = \log T$.

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